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The effect of consumption of animal milk compared to infant formula for non-breastfed/mixed-fed infants 6-11 months of age. A systematic review (protocol)

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The effect of consumption of animal milk compared to infant formula for non-breastfed/mixed-fed infants 6-11 months of age. A systematic review (protocol)

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The effect of consumption of animal milk compared to infant formula for non-breastfed/mixed-fed infants 6-11 months of age. A systematic review (protocol)

Aamer Imdad, Julie Ehrlich, Joseph Catania, Emily Tanner Smith, Abigail Smith, Olivia Tsistinas, Zulfiqar Ahmed Bhutta,

Abstract

Introduction

Prevalence rates of breastfeeding remain low even though the World Health Organization and American Academy of Pediatrics recommend exclusive breastfeeding for the first 6 months of life in combination with appropriate complementary feeding beyond six 6 months of age. There have been several studies that address the implication of drinking animal milk and/or infant formula on children's health and development when breastfeeding is not offered during the first year of life. Vast improvements have been made in infant formula design, which may increase its benefits compared to animal's milk. The objective of this review is therefore to synthesize the most recent evidence on the effects of the consumption of animal milk compared to infant formula in non-breastfed or mixed breastfed infants aged 6-11 months.

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Methods and Analysis

We will conduct a systematic review and meta-analysis of studies that assessed the effect of animal milk as compared to formula or mixed-fed (breastmilk and formula) on infants aged 6-11 months. We will conduct systematic electronic searches on multiple databases. The primary outcomes of interest include anemia, gastrointestinal blood loss, weight for age, height for age, and weight for height. We will include randomized and non-randomized studies with a control group. We will use the Cochrane risk of bias tools to assess the risk of bias in the included studies. We will use meta-analysis to pool findings if the identified studies are conceptually homogenous and data are available from more than one study. We will pool dichotomous outcomes using relative risks and will pool continuous outcomes using mean difference effect sizes; all mean effect size estimates will be reported with their corresponding 95 % confidence intervals. We will assess the overall quality of evidence using the GRADE approach.

Ethics and Dissemination

This review will contribute significantly to existing literature on infant feeding. This review is being conducted as part of World Health Organization call for systematic reviews on infant and young feeding. The findings from this meta-analysis will be used to guide recommendations on whether the use of infant formula is better than animal's milk in infants aged 6-11 months.

Registration

Registered to PROSPERO on September 23rd, 2020

https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020154352

Strengths and Limitations of the Study

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- The findings from this systematic review will be used to guide recommendations on whether the use of infant formula is better than animal's milk in infants aged 6-11 months.
- Our primary focus is to find randomized trials on this subject however infant nutrition is under researched and clinical trials are difficult to conduct in this age group due to ethical reasons.
- We might find evidence from randomized and non-randomized studies and that may pose a challenge in combining the data in a meta-analysis and grading the overall quality of evidence.

Keywords: Animal's milk, infant formula, anemia

Introduction

The World Health Organization and American Academy of Pediatrics recommend exclusive breastfeeding for the first 6 months of life and continuation of breastfeeding beyond 6 months in combination with appropriate complementary feedings for up to 2 years or beyond[1,2]. However, prevalence rates of breastfeeding still remain low in many contexts, with many infants still being given formula milk or cow's milk before or after 6 months of age[3,4]. Use of cow's milk in infancy has been associated with gastrointestinal blood loss, iron deficiency anemia, and increased solute load for kidneys[5-8]. The composition of cow's milk is vastly different from that of breastmilk and infant formula, which is manufactured to mimic breastmilk[9]. Cow's milk has a high amount of protein (3.4 g/100 ml vs .7-1.4100 ml in breastmilk)[10], and the composition of the protein is different—50-80% of breastmilk is comprised of whey protein versus only 18% for cow's milk[9]. Ingesting a large amount of protein in the first year of life has been associated with

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childhood obesity[6,9]. Unmodified goat's milk also contains high amounts of protein and mineral content and low folate content. Goat's milk does not appear to have any clear nutritional advantage over cow's milk and is not less allergenic[11]. There have been several studies that have examined the implications of drinking animal milk and/or infant formula on children's health and development[5,7,12-17]. The past two decades have also yielded vast improvements in the design of infant formulas, which may increase the benefits of infant formula over animal milk[18]. The objective of the proposed review is, therefore, to synthesize the most recent research on the effects of the consumption of animal milk compared to infant formula in non-breastfed or mixed breastfed infants aged 6-11 months of age.

Objective

For non-breastfed or mixed-fed (breastmilk and formula) infants 6-11 months of age, is the consumption of animal milk compared to infant formula, associated with beneficial or adverse outcomes for health and development?

Methods and Analysis

Study type: We will include individual and cluster randomized trials. We will also include quasi-randomized experimental design studies, defined as those that attempted to randomly allocate participants to conditions but the method of randomization was not truly random[19]. For instance, an example of a quasi-randomized design is an experimental study where the randomization was attempted but the methods of sequence generation were inadequate such as randomization based on date of birth of participants, date of enrollment in the study. We will also include cohort studies, defined as a study on a group of participants based on certain criteria (e.g., geography,

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age, etc.) but none of the participants have the outcome at the time of recruitment and develop outcomes over time in relation to exposure vs. no exposure to a risk factor. A cohort study could be prospective or retrospective[19]. We will exclude case control studies, case series, case reports and cross-sectional studies. All analyses will be conducted and reported separately for randomized and non-randomized studies.

Population: We will include studies that included apparently healthy infants 6-11 months of age who were non-breastfed or mixed fed (breastmilk and formula) irrespective of gestational age and birthweight. We will include studies that started animal milk feeding between 4 months and 11 months of age. If a study includes participants who began intervention before 4 months of age but includes participants older than 4 months of age as well, we will consider that study eligible if more than half (50%) of the study participants meet the inclusion criteria (i.e. 4-11 months of age). We will include studies where participants are diagnosed with HIV or have had HIV exposure. We will exclude studies if participants are diagnosed with AIDS. We will exclude studies that focused on breastfed infants only without other types of milk provided. We will exclude studies with participants who have chronic diseases such as bronchopulmonary dysplasia, genetic disorders, aerodigestive problems, or congenital anomalies.

***Intervention*:** We will include studies that assessed the effect of animal milk in infants 6-11 months of age. The animal milk could be cow milk, goat milk, or camel milk. We will include studies in which animal milk was the main milk drink as defined by study authors or more than 50% of the infant's milk intake was animal milk. We will include studies irrespective if the animal milk was boiled or not boiled, pasteurized or unpasteurized, or if the animal milk was

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full-fat, reduced fat, or skim milk. We will exclude studies that used plant-based milk such as rice milk, pea milk, coconut milk, or almond milk.

***Comparison*:** The comparison group in the included studies will be formula feeding or mixed feeding (i.e., breastfeeding and formula feeding). We will include studies irrespective of the type of formula used; namely, this could include cow's milk-based formula, partially or extensively hydrolyzed formula, or plant-based formulas such as soy formula. The Federal Food, Drug, and Cosmetic Act (FFDCA) defines infant formula as "a food which purports to be or is represented for special dietary use solely as a food for infants by reason of its simulation of human milk or its suitability as a complete or partial substitute for human milk"[20].

Outcomes:

Primary outcomes

-Anemia (dichotomous outcome, as defined by authors) at 7 months, 9 months, 12 months, and the longest follow up

-Gastrointestinal blood loss (dichotomous outcome based on stool occult testing) at 7 months, 9 months, 12 months, and the longest follow up

-Weight for age (kg or Z scores) at 7 months, 9 months, 12 months, and the longest follow up

-Height for age (cm or Z scores) at 7 months, 9 months, 12 months, and the longest follow up

-Weight for height Z score at 7 months, 9 months, 12 months, and the longest follow up

Secondary outcomes

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-Iron deficiency anemia (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Blood iron level (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Blood ferritin level (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Hemoglobin concentration in the stool (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Hemoglobin concentration in the serum (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Fatty acid status (continuous outcome) at 7 months, 9 months, 12 months and longest follow up

-Gut health: Diarrhea (> 3 loose stools per day) (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Gut health: Constipation (<3 bowel movements per week) (dichotomous outcome) at 7 months, 9 months, 12 months and longest follow up

-Pneumonia (as defined by authors) (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Allergy (IgE-Mediated and non-IgE Mediated and mixed) (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

We will include both immunoglobulin E (IgE) mediated, non-IgE mediated and mixed milk allergies. IgE mediated can be identified with a skin prick test or blood test for specific IgE and symptoms typically arise immediately after ingesting the milk product. Non-IgE mediated allergies will not be identified with a skin prick test or blood test and symptoms typically arise

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48 hours after ingestion of milk product [21]. Non-IgE mediated allergies can be identified using allergen elimination and standard challenge procedure in which the allergen is removed from the diet for up to 2 weeks and then reintroduced to determine if symptoms are elicited[22].

- Obesity (dichotomous outcome) at 1 year and longest follow up

Obesity is defined as a BMI at or above the 95th percentile for children and teens of the same age and sex[23]. It can also be defined as BMIZ>3 for children under 5 years of age.

-Overweight (dichotomous outcome) at 1 year and the longest follow up

Overweight is defined as a BMI at or above the 85th percentile and below the 95th percentile for children and teens of the same age and sex[23]. It can also be defined as BMIZ>2 for children under 5 years of age.

-Neurodevelopmental outcomes (continuous outcome) at 1 year and the longest follow up

The term neurodevelopment is a composite term that refers to cognitive, neurologic, and/or sensory outcomes. This may include intellectual disability as measured on the Mental Developmental Index of the Bayley Scales of Infant Development, gross motor delay measured on Gross Motor Function Classification System, etc.

If a study does not report outcomes at 7 months but instead reports outcomes at 6 months or 8 months, it will be included with outcomes at 7 months. Similarly, if a study does not report outcomes at 9 months but instead reports outcomes at 10 months, it will be included with outcomes at 9 months. If a study does not report outcomes at 12 months, but instead reports those outcomes at 11, 13, or 14 months, it will be included with outcomes at 12 months.

Literature Search:

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We will conduct systematic electronic searches on multiple databases, including PubMed, EMBASE, the Cochrane Central Register for Controlled Trials, Web of Science, CINHAL, Scopus, WHO Global Index Medicus. There will be no restrictions applied to the searches based on outcomes, study design, publication status, publication date, or language. A proposed search strategy for PubMed is shown in Appendix 1. We will search ClinicalTrials.gov to identify ongoing studies. We will also search the websites of relevant international agencies, such as the WHO (including WHO's Reproductive Health Library), UNICEF, Global Alliance for Improved Nutrition, International Food Policy Research Institute, International Initiative for Impact Evaluation (3ie), Nutrition International, World Bank, USAID and affiliates (e.g., FANTA, SPRING), and the World Food Program. The reference sections of previously published reviews and the latest published studies will also be searched for potentially eligible studies.

Data Extraction and Synthesis:

Selection of studies

Searches from all the databases will be combined in bibliographic software (EndNote) and duplicates will be removed. Two authors (AI and JC or AI and JE) will first screen the titles to identify potentially eligible studies; any studies retained at this step will proceed to the second stage of screening for a full-text review; finally, any studies deemed eligible during the full-text review will proceed to the third step of full data extraction. Any conflict will be resolve by discussion and with the help of the senior author on the team if needed. We will use coding software (Covidence) to conduct the screening. If a study is only available in abstract form, we will write to authors to obtain details on methods and results. If a study is available in a language other than English, we will attempt to have the translation completed using the local resources. If

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a study was published in more than one report (multiple publications), we will count those multiple reports as a single study and extract information from all the available reports as needed.

Data Extraction

A data extraction sheet will be designed and piloted for the extraction of information from the selected studies. Two authors (JE and JC) will independently extract the data and compare their findings. Any conflict will be resolved by discussion and with the help of the senior author on the team if needed. We will extract the information on type of study (randomized control trial, quasi-randomized experimental design, or cohort study) study site, study year, study population (age, nutritional status, percent female), methods for data collection and analysis, intervention (dose, duration, frequency), age of outcome assessment, comparison, outcomes, whether the results were adjusted for confounders, what confounders were considered (socioeconomic status, maternal employment, maternal age, maternal level of education, age-appropriate complementary feeding) and risk of bias (supplementary document).

Assessment of risk of bias in included studies

Study risk of bias will be assessed using the Cochrane risk of bias tool (ROB 2.0)[24] for randomized controlled trials and using the Cochrane risk of bias in non-randomized studies (ROBINS-I) tool for non-randomized studies of interventions[25]. Two review authors will independently evaluate and agree on the risk of bias for the individual studies for an outcome. Any disagreements will be resolved by discussion and if no agreement can be made, the senior review author will be consulted. We will assess the effect of assignment to intervention (the

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intention-to-treat effect) by addressing five domains of signaling questions including: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result. Each domain will get a ranking of low risk of bias, some concerns of bias, or high risk of bias. We will include quotes from the study for each signaling question as evidence for our ranking decision. The overall risk of bias will be determined based on the worst ranking for individual domains. For example, if only one domain is ranked "some concerns" then the overall risk of bias will be "some concerns".

Data synthesis

We will develop a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram based on the search strategy and eligibility assessment to show the flow of included and excluded studies[26]. We will analyze the randomized and non-randomized studies separately. We will report findings from all included studies in a narrative synthesis and will also conduct meta-analyses to synthesize evidence across studies quantitatively. Meta-analyses will be conducted when data are available from more than one study, and when there is clinical and methodological homogeneity in the included studies. Dichotomous outcomes will be measured using relative risk effect sizes and will be reported with their corresponding 95 % confidence intervals. Continuous outcomes will be measured using mean difference effect sizes and will also be reported along with their 95 % confidence intervals. We will use the generic inverse variance weighting method for meta-analysis. We will use the random effects model for meta-analysis given that there might be heterogeneity in effects due to variability in the study populations and interventions used. We will use RevMan[27] and Stata[28] software for all statistical analysis.

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In the case of morbidity outcomes, we will combine all available data whenever possible if outcomes are measured in different ways. For example, we will include all types of diarrhea (mild, moderate and severe) as a dichotomous value (yes/no) if participants had greater than 3 instances of loose stools per day. We will include occurrence of anemia, gastrointestinal blood loss based on stool occult testing, iron deficiency anemia, pneumonia, and allergies (non-IgE, IgE mediated) throughout the study as dichotomous values (yes/no). For dichotomous outcomes, we will extract the total number of participants in each group and the number of participants experiencing an event. We will pool the dichotomous outcomes in a meta-analysis to obtain a summary estimate in the form of relative risk and report it with its corresponding 95 % confidence interval (CI). We will include hemoglobin concentration in stool, weight for age, height for age, and weight for height as continuous values. For continuous outcomes, we will pool the data to obtain a pool mean difference and report it with its 95 % CI. In the event that data are reported in different units (e.g. few studies report weight in kg and the others report in Z scores), we will use a standardized mean difference effect size and report it with its 95 % CI.

Unit of analysis issues

Multiple-arm trial

Studies with multiple treatment arms will be included if eligible. For multiple-arm trials we will include data in a way that the only difference between the groups is use of animal milk.

Cluster trials

Cluster assignment trials will be analyzed together with individual randomized trials. We will use the cluster adjusted values; If the trial results are not adjusted for cluster design, we will adjust the result by methods given in the Cochrane handbook for systematic reviews[19].

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Studies with missing data

We will document attrition during data extraction. If the authors report the missing data and report the imputations for the missing data, we will use the latter. If data are missing for some cases, or if reasons for dropout are not reported, we will contact the trial authors to request the full data. In the event that a study does not report the SD for the continuous outcome, we will write to authors to request the data. If the SD data are not available from authors, we will use SD from a similar study that has similar study population. We will prefer to use the final values of a continuous outcome for a given follow up. If the final values are not available but the difference between the end and the start of the study, we will write to authors to request the final values. If the final values are not available, we will use the difference or rate of change.

Assessment of heterogeneity

Assessment of heterogeneity in meta-analysis determines how different the included studies are from each other. Clinical heterogeneity refers to differences in patient population or treatment whereas methodological heterogeneity refers to differences in how the study was designed. Statistical heterogeneity refers to large differences in the outcome (outside of chance) that may be due to clinical or methodological heterogeneity[29]. Statistical heterogeneity will be assessed using Tau^2, χ^2 and I^2 statistics. We will assess statistical heterogeneity by visual inspection of forest plots, by performing the Chi^2 test (assessing the P value) and by calculating the Tau^2 and I^2 statistic. Statistical heterogeneity will be considered substantial if the P value is less than 0.10, I^2 value exceeds 50%, and inspection of forest plots show substantial variability in the effect of

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the intervention. We will perform subgroup analysis to determine the reasons for any identified statistical heterogeneity.

Assessment of reporting bias

Small study and publication bias will be assessed using funnel plots and regression tests for funnel plot asymmetry when the meta-analysis includes at least ten studies.

Subgroup analysis and investigation of heterogeneity

-Age group: 7 months vs. 9 months

If a study does not begin intervention at 7 months but instead at 6 months or 8 months, it will be included in the 7 months age sub-group. Similarly, if a study does not begin intervention at 9 months but instead at 10 months, it will be included with outcomes at 9 months.

-Country: Low and middle-income country vs. high-income country

-Type of Feeding: Non-breastfeeding vs. mixed feeding

- Type of Animal Milk: Cow, goat, buffalo, camel, sheep

We will analyze subgroups using chi-square test.

Sensitivity analysis

-Studies with a high overall risk of bias excluded

-Random vs. fixed effect meta-analysis model

Rating of overall quality of evidence

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We will assess the overall quality of evidence for the effect of the intervention on each primary outcome using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) using the software GradePro. The GRADE approach entails identifying and assessing many different features that affect the certainty of evidence in the review, including type of study design, statistical heterogeneity, directness of evidence, within-study risk of bias, risk of publication bias, and precision of effect estimates.[30] We will consider evidence from randomized trials as high quality and downgrade the evidence one level for serious limitations (table). We will rate the overall quality of the body of evidence as very low (we have very little confidence in the effect estimate), low (we have limited confidence in the effect estimate), moderate (we have moderate confidence in the effect estimate; the true effect is likely close to the estimate of the effect), or high (we have high confidence that the true effect lies close to that of the estimate of the effect). We will present the results of the GRADE assessment in the form of a summary of findings tables for the primary outcomes.

Amendments

We will do the literature searches, screening of titles, selection of studies, data extraction, and analysis according to a priori plan described in this protocol. If we do any additional analysis or change any of the a priori strategies, we will clearly describe that in the methods section.

Patient and Public Involvement

No patient or public involvement was considered in preparation of this protocol.

List of abbreviations

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World Health Organization (WHO)

The Federal Food, Drug, and Cosmetic Act (FFDCA)

Confidence Interval (CI)

Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

Grading of Recommendations Assessment, Development, and Evaluation (GRADE)

Ethics and dissemination

Ethics approval and consent to participate

This is a systematic review and there will be no direct contact with the patients. This study therefore does not involve human subjects research and does not need IRB approval.

Consent for publication

Authors give consent for publication of this review

Availability of data and materials

We will keep all the data available for review by the editors and peer reviewers and to general public on request

Authors' contributions

AI and JE wrote the first draft of the manuscript. AS developed the search strategy. ETS provided the support for methodology. ZAB provided the support for the content being evaluated in the review. All authors edited and reviewed the manuscript.

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Competing interests

Authors declare they do not have any conflicts of interest to disclose.

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Table: ***GRADE method for rating the quality of evidence***

Study Design	Quality of Evidence	Lower if	Higher if
Randomized trial	High	Risk of Bias -1 Serious -2 Very serious Inconsistency -1 Serious -2 Very serious Indirectness -1 Serious -2 Very serious Imprecision -1 Serious -2 Very serious Publication bias -1 Serious -2 Very serious	Large effect + 1 Serious + 2 Very serious Dose response + 1 Evidence of a gradient All plausible confounding + 1 Would reduce a demonstrated effect or + 1 Would suggest a spurious effect when results show no effect
Quasi-Randomized trial	Moderate		
	Low		
	Very low		

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Appendix 1:

PubMed

((("cow milk*"[tiab] OR "cow's milk*"[tiab] OR "cows milk*"[tiab] OR "bovine milk*"[tiab] OR "bovine's milk*"[tiab] OR "goat milk*"[tiab] OR "goat's milk*"[tiab] OR "goats milk*"[tiab] OR "caprine milk*"[tiab] OR "buffalo milk*"[tiab] OR "buffalo's milk*"[tiab] OR "camel milk*"[tiab] OR "camel's milk*"[tiab]OR "camels milk*"[tiab] OR "sheep milk*"[tiab] OR "sheep's milk*"[tiab] OR "ewe milk*"[tiab] OR "ewe's milk*"[tiab] OR "ewes milk*"[tiab] OR "ovine milk*"[tiab]) AND ("Infant Formula"[Mesh] OR formula*[tiab])) AND ("Infant"[Mesh] OR infant*[tiab] OR infancy[tiab] OR baby[tiab] OR babies[tiab] OR neonat*[tiab] OR "neo nat*"[tiab] OR newborn*[tiab] OR "new born*"[tiab] OR "newly born*"[tiab] OR "Child"[Mesh] OR child*[tiab] OR youth[tiab] OR juvenile*[tiab])) NOT ("Animals"[Mesh] NOT ("Animals"[Mesh] AND "Humans"[Mesh]))

CINAHL

(TI ("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel

milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*") OR AB ("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*"OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*"))
AND
((MH "Infant Formula") OR (TI formula* OR AB formula*))
AND
(((MH "Child") OR (MH "Infant+")) OR (TI (infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*) OR AB (infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)))
NOT
(((MH "Animals+") OR (MH "Animal Studies") OR (TI "animal model*")) NOT (MH "human"))
Exclude Medline Records

SCOPUS

(TITLE-ABS ("cow milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR OR "ewes milk*" OR "ovine milk*")) AND (TITLE-ABS (formula*)) AND (TITLE-ABS (infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)) AND NOT INDEX (medline)

Embase

1 'cow milk'/exp OR 'goat milk'/exp OR 'buffalo milk'/exp OR 'camel milk'/exp OR 'sheep milk'/exp
2 'cow milk*':ti,ab OR 'cows milk*':ti,ab OR 'cow s milk*':ti,ab OR 'bovine milk*':ti,ab OR 'bovine s milk*':ti,ab OR 'goat milk*':ti,ab OR 'goat s milk*':ti,ab OR 'goats milk*':ti,ab OR 'caprine milk*':ti,ab OR 'buffalo milk*':ti,ab OR 'buffalo s milk*':ti,ab OR 'camel milk*':ti,ab OR 'camel s milk*':ti,ab OR 'camels milk*':ti,ab OR 'sheep milk*':ti,ab OR 'sheep s milk*':ti,ab OR 'ewe milk*':ti,ab OR 'ewes milk*':ti,ab OR 'ewe s milk*':ti,ab OR 'ovine milk*':ti,ab
3 #1 OR #2
4 'artificial milk'/exp
5 formula*:ti,ab
6 #4 OR #5
7 'infant'/exp OR 'infancy'/exp OR 'baby'/exp OR 'newborn'/exp OR 'child'/exp OR 'juvenile'/exp
8 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR neonat*:ti,ab OR 'neo nat*':ti,ab OR newborn*:ti,ab OR 'new born*':ti,ab OR 'newly born*':ti,ab OR child*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
9 #7 OR #8
10 #3 AND #6 AND #9
11 #10 NOT ([animals]/lim NOT [humans]/lim)
12 #11 NOT [medline]/lim

Cochrane CENTRAL

1 "cow milk*":ti,ab OR "cows milk*":ti,ab OR "cow's milk*":ti,ab OR "bovine milk*":ti,ab OR "bovine's milk*":ti,ab OR "goat milk*":ti,ab OR "goat's milk*":ti,ab OR "goats milk*":ti,ab OR "caprine milk*":ti,ab OR "buffalo milk*":ti,ab OR "buffalo's milk*":ti,ab OR "camel milk*":ti,ab OR "camel's milk*":ti,ab OR "camels milk*":ti,ab OR "sheep milk*":ti,ab OR "sheep's milk*":ti,ab OR "ewe milk*":ti,ab OR "ewes milk*":ti,ab OR "ewe's milk*":ti,ab OR "ovine milk*":ti,ab
2 MeSH descriptor: [Infant Formula] explode all trees
3 formula*:ti,ab
4 #2 OR #3
5 MeSH descriptor: [Child] explode all trees
6 MeSH descriptor: [Infant] explode all trees
7 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR neonat*:ti,ab OR "neo nat*":ti,ab OR newborn*:ti,ab OR "new born*":ti,ab OR "newly born*":ti,ab OR child*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
8 #5 OR #6 OR #7

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9 MeSH descriptor: [Animals] explode all trees
10 MeSH descriptor: [Humans] explode all trees
11 (#9 NOT(#9 AND #10))
12 #1 AND #4 AND #8 NOT #11
13 "accession number" near pubmed
14 #12 NOT #13

LILACS

ti:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*") OR (ab:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*")) AND (((ti:(formula*)) OR (ab:(formula*))) OR ((mh:("infant formula")))) AND (((ti:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)) OR (ab:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*))) OR ((mh:("infant")) OR ((mh:("child"))))) AND NOT ((mh:("animals")) AND NOT ((mh:("animals")) AND ((mh:("humans")))))

Global Index Medicus

ti:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*") OR (ab:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*")) AND (((ti:(formula*)) OR (ab:(formula*))) OR ((mh:("infant formula")))) AND (((ti:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)) OR (ab:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*))) OR ((mh:("infant")) OR ((mh:("child"))))) AND NOT ((mh:("animals")) AND NOT ((mh:("animals")) AND ((mh:("humans")))))AND (db:("WPRIM" OR "IMEMR" OR "IMSEAR" OR "AIM"))

Web of Science

1 TI=("cow* milk*" OR "bovine* milk*" OR "goat* milk*" OR "caprine milk*" OR "buffalo* milk*" OR "camel* milk*" OR "sheep* milk*" OR "ewe* milk*" OR "ovine milk*")
2 AB=("cow* milk*" OR "bovine* milk*" OR "goat* milk*" OR "caprine milk*" OR "buffalo* milk*" OR "camel* milk*" OR "sheep* milk*" OR "ewe* milk*" OR "ovine milk*")
3 #2 OR #1
4 TI=(formula*)
5 AB=(formula*)
6 #5 OR #4
7 TI=(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new* born*" OR child* OR youth OR juvenile*)
8
9 #8 OR #7

10 #9 AND #6 AND #3

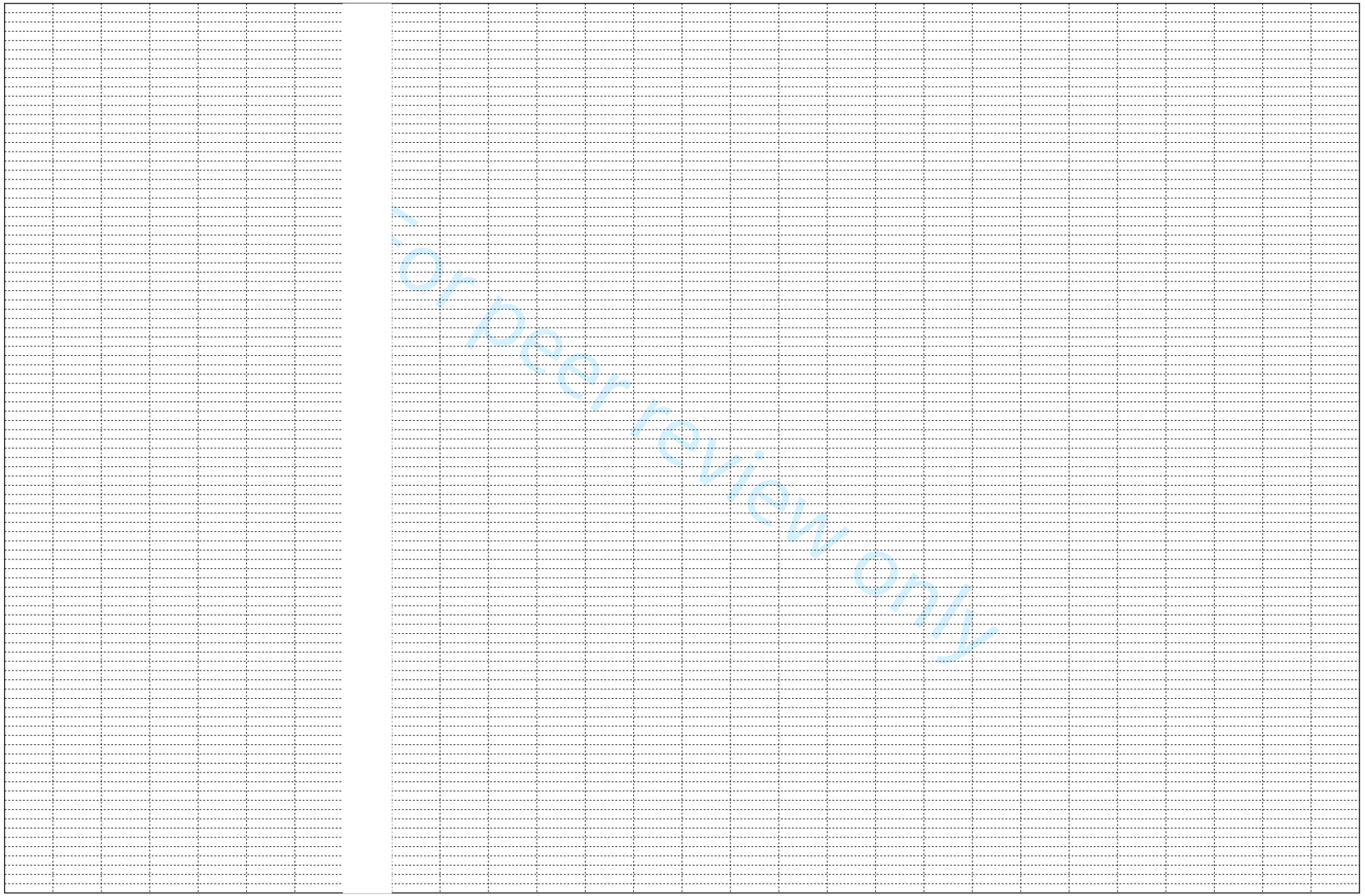
11 #9 AND #6 AND #3 Exclude Medline

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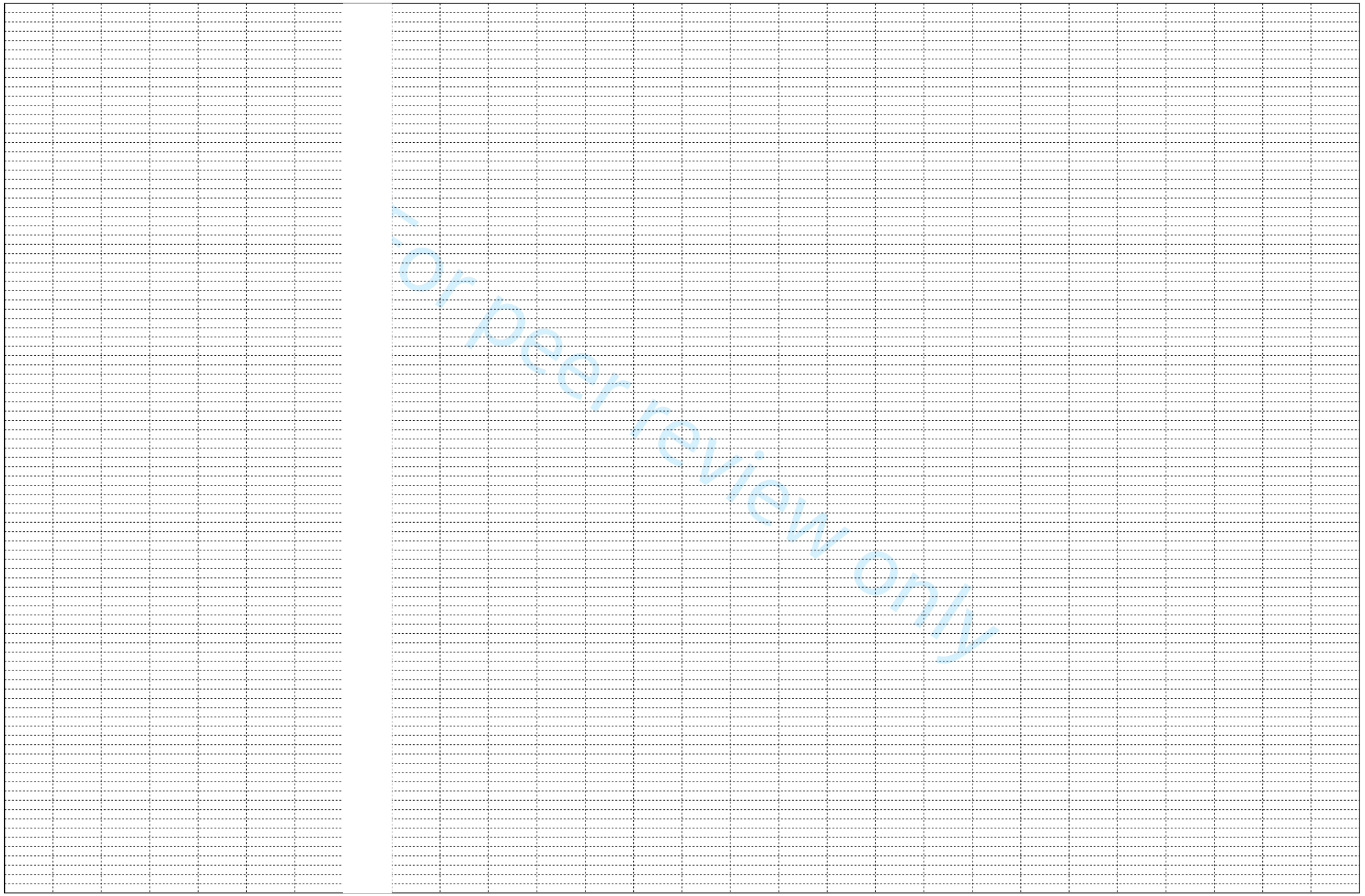
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Author	Publication year	Name of data extractor	Form: conference abstract only (1), journal publication (2),	Study Design	Cluster Design Trial: no (0), yes (1)	Study Setting			Population									Characteristic of Intervention			Characteristic of comparison			Study confounders		Notes	Minimal loss to follow up? (more than 20% is serious limitation)
						Country	LMIC (1) vs. HIC (2) -->	Year(s) of Data Collection	Description of Participants		Age initiation of animal milk	Nutritional Status: healthy	% Female (N) (Give for total	Were there more than one	Total number of eligible	Total number of eligible	Number of Participants	Type of Animal milk (eg: cow	Frequency of animal milk	Volume of animal milk	breastmilk and formula	Volume	Frequency	Were results adjusted for	What confounders		
									Inclusion	Exclusion																

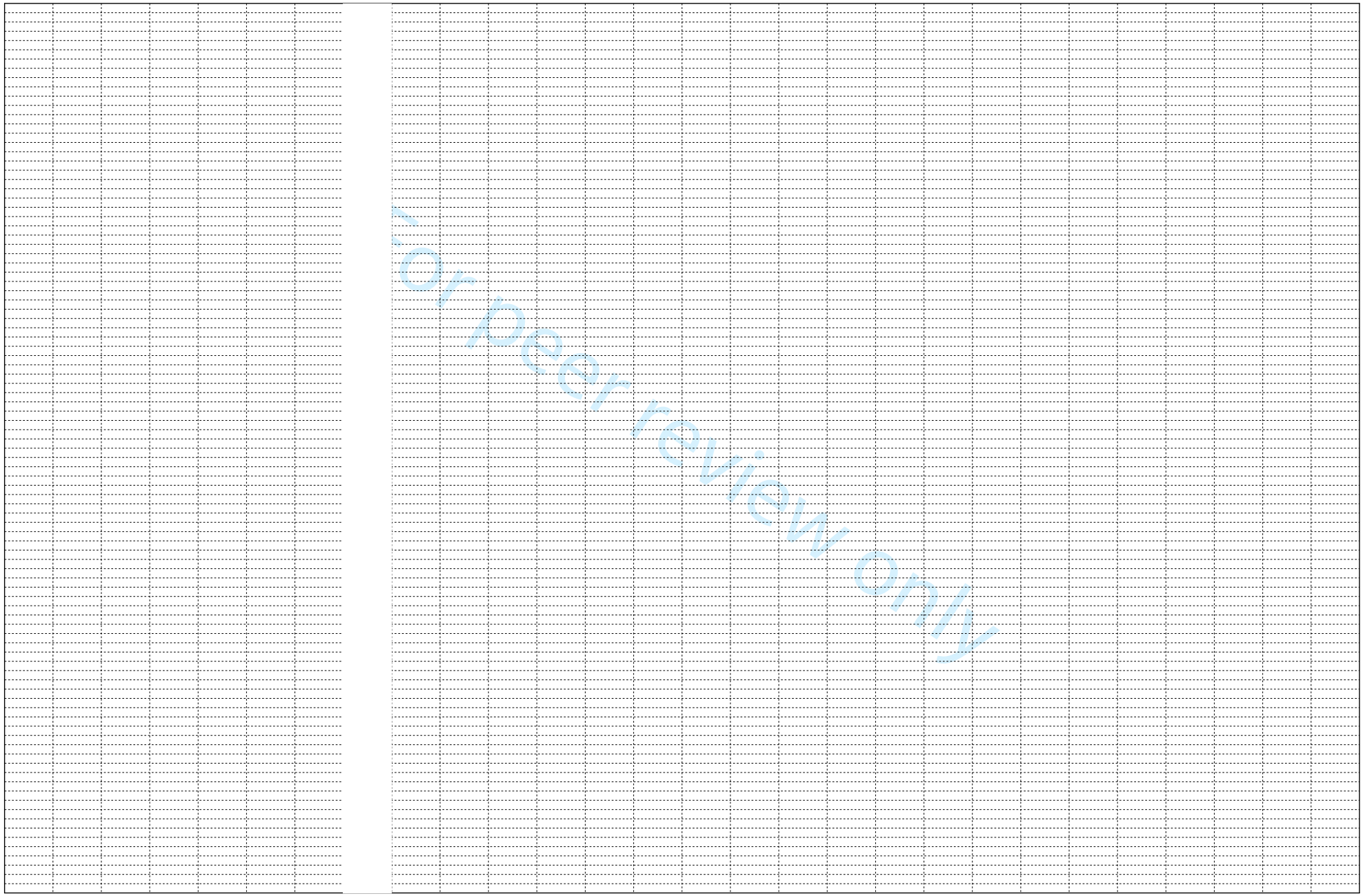
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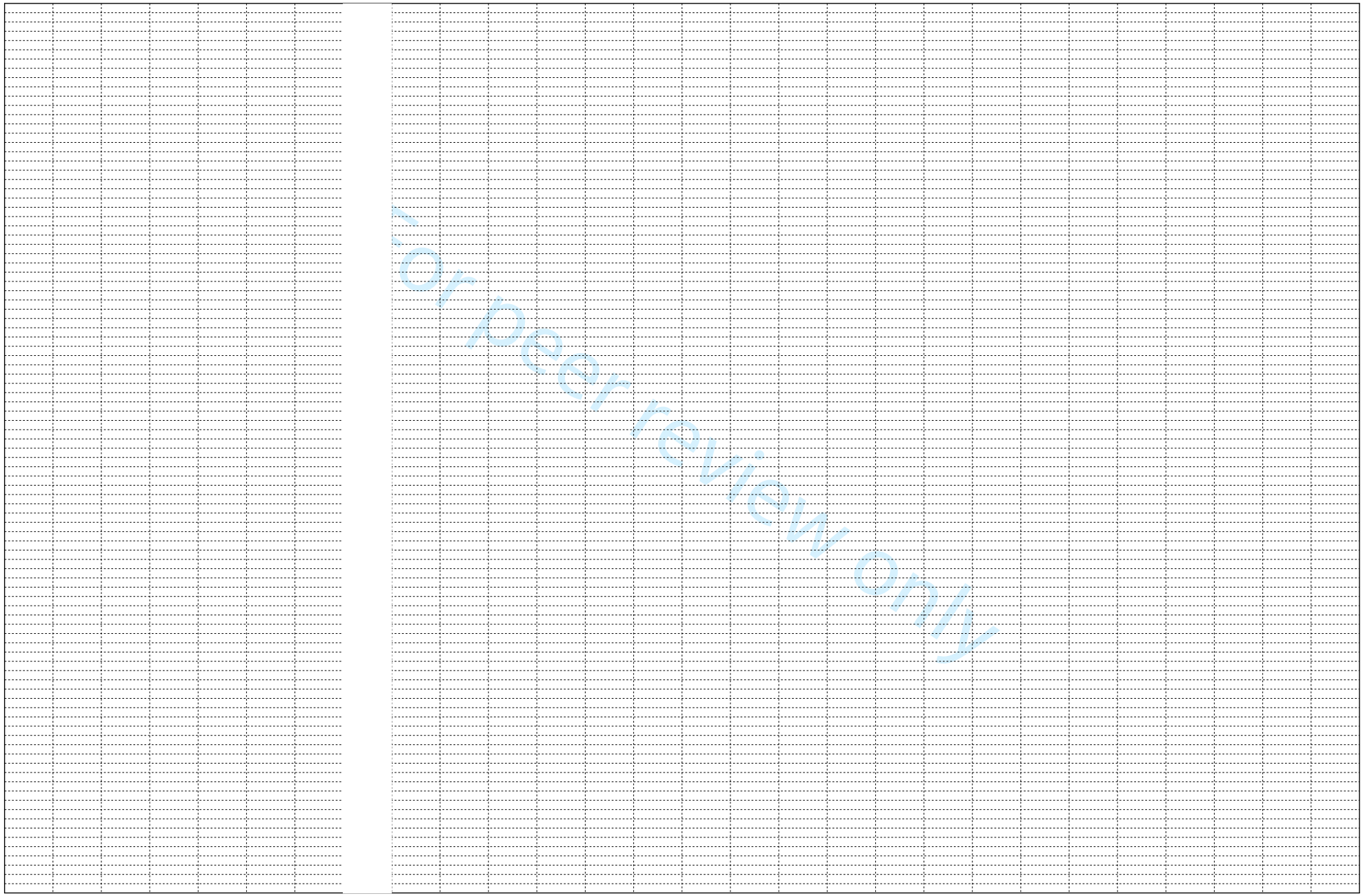
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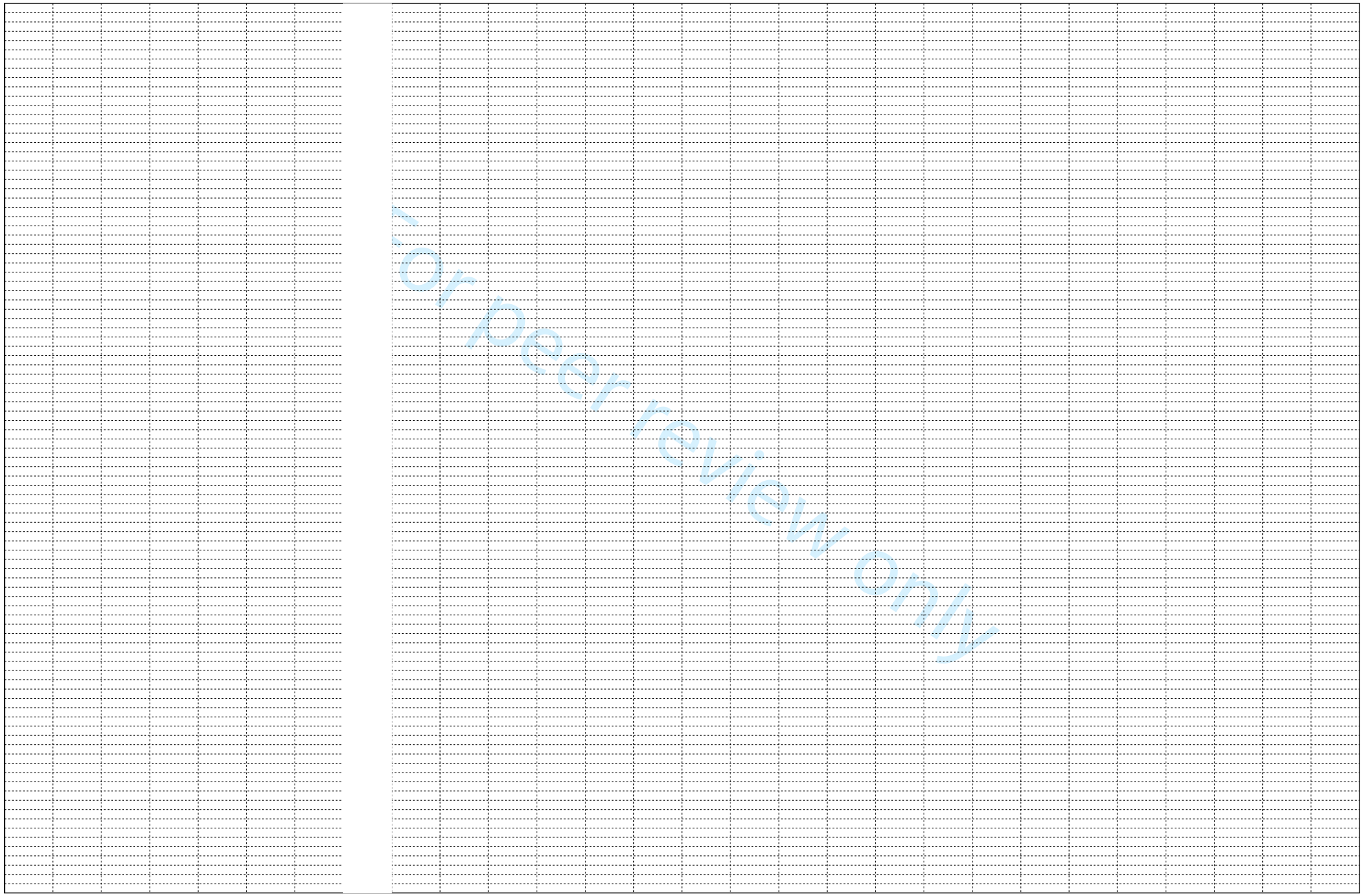
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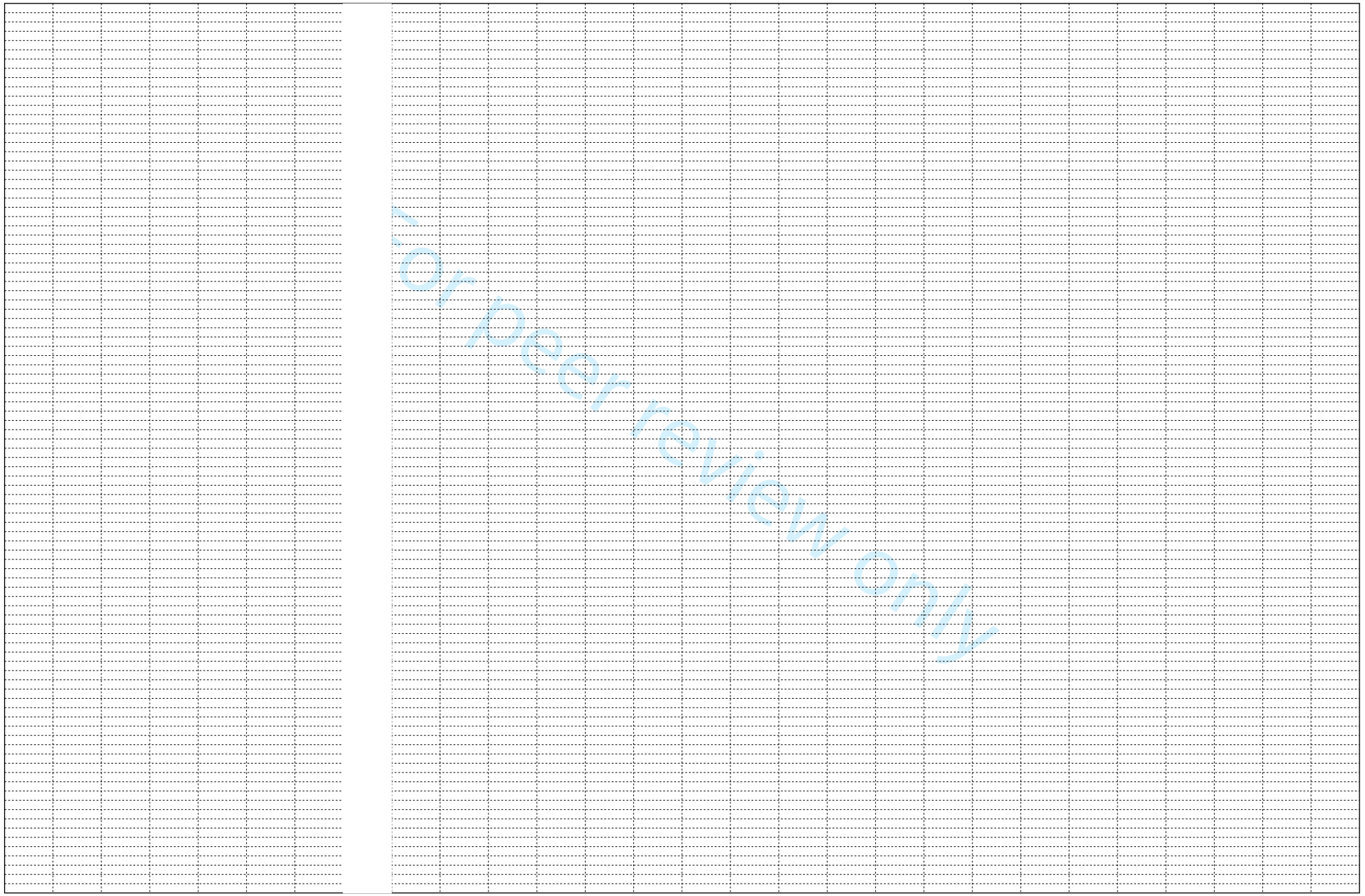
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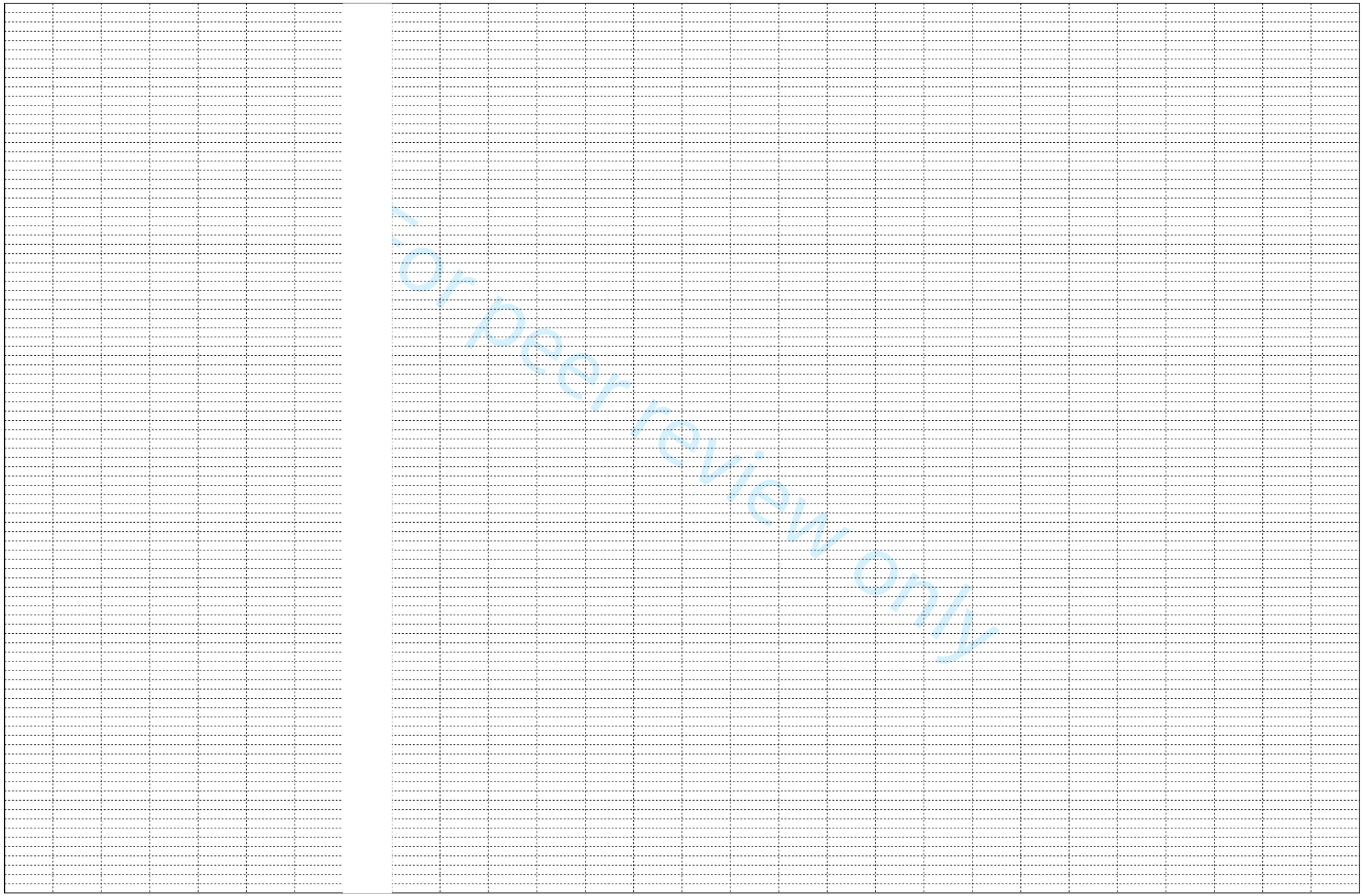
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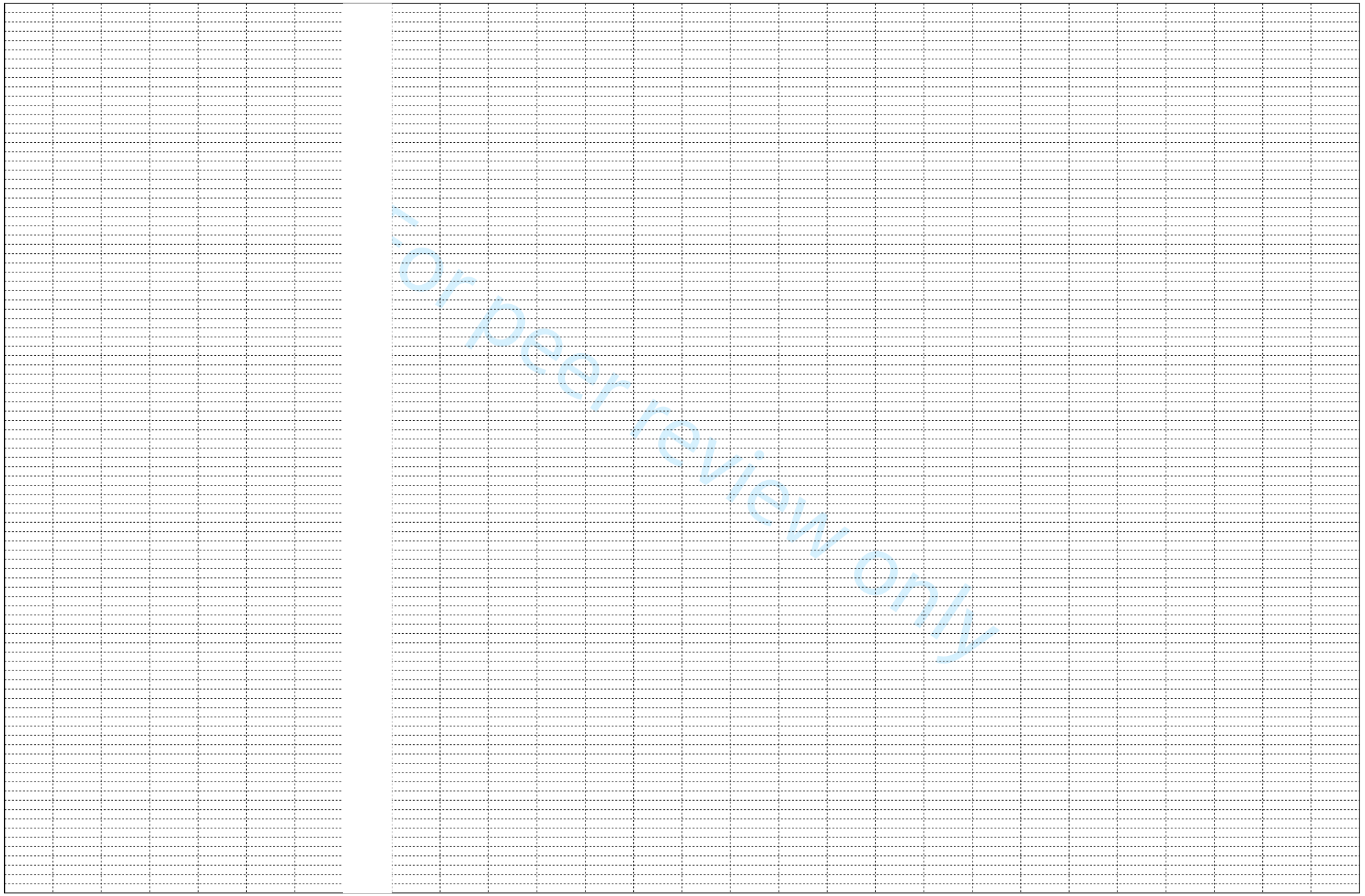
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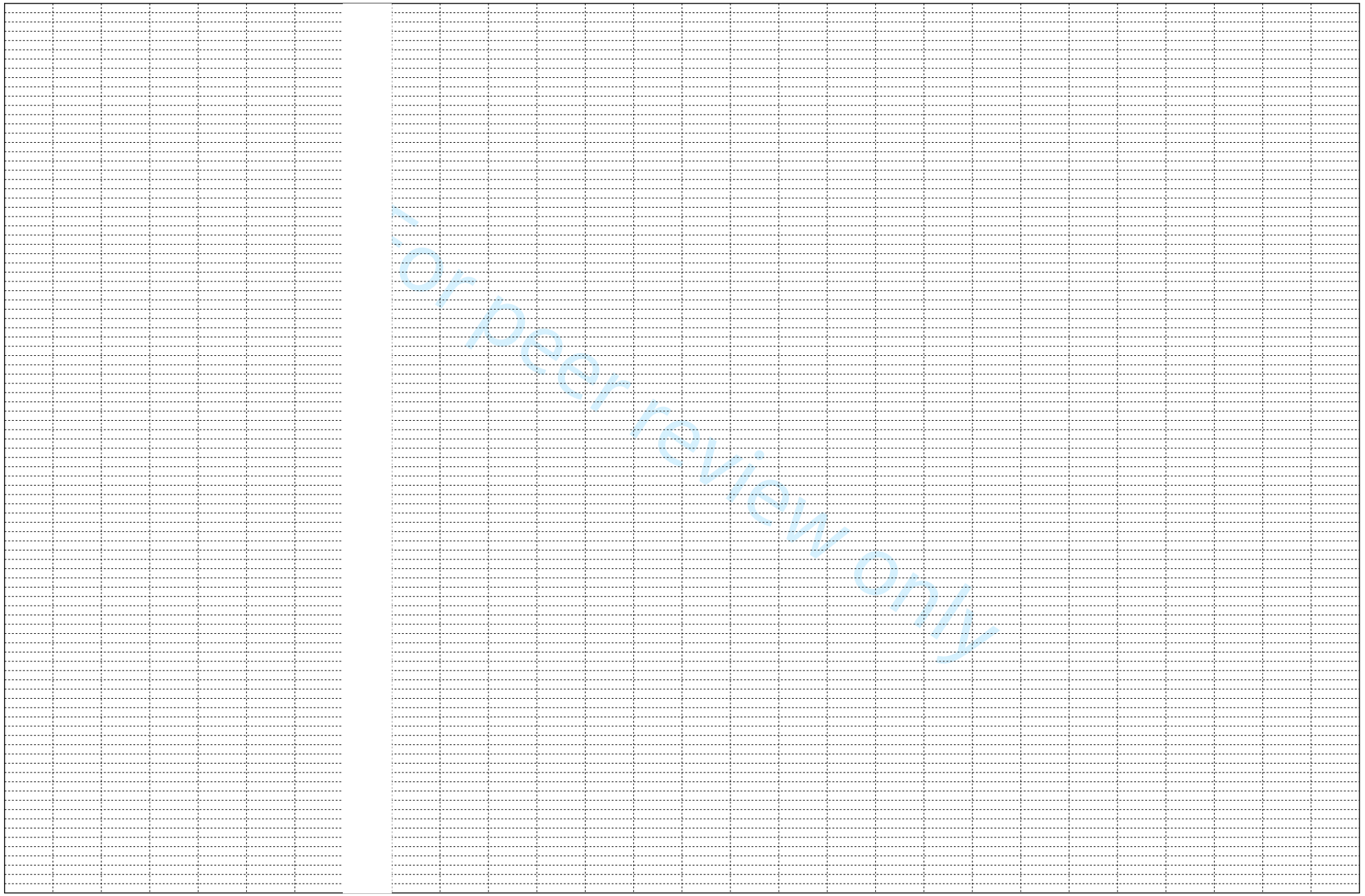
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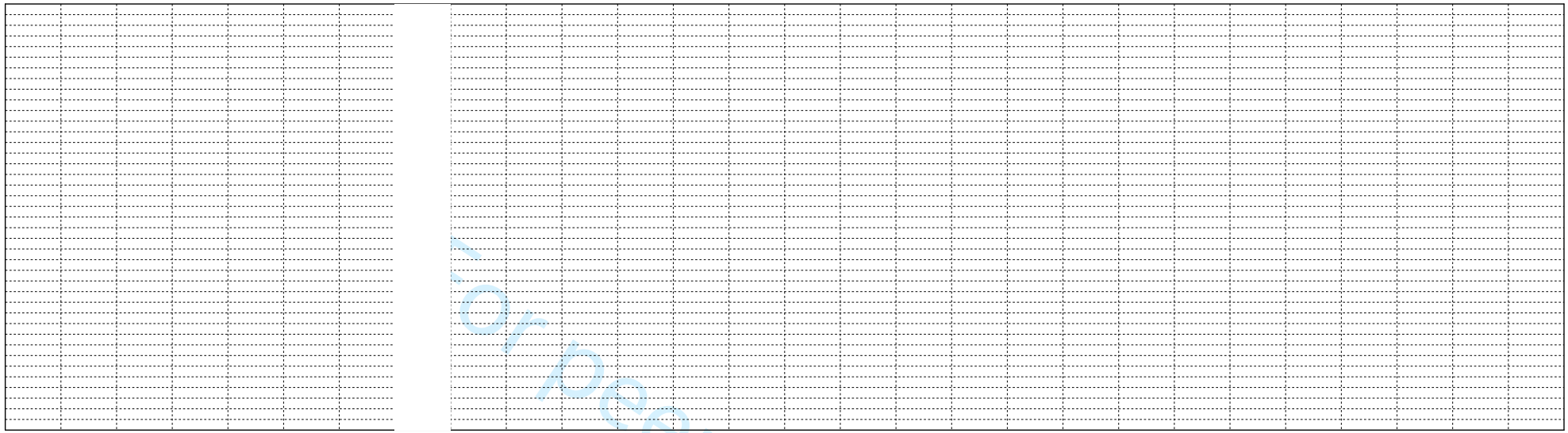
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For peer review only



For peer review only



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year)	Anemia at 7 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Anemia at 9 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Anemia at 12 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Anemia at longest follow-up (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Gastrointestinal blood loss at longest follow-up (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Gastrointestinal blood loss at 7 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Gastrointestinal blood loss at 9 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Gastrointestinal blood loss at 12 months (yes/no)				
	events in	intervention	events in control	the control group	notes

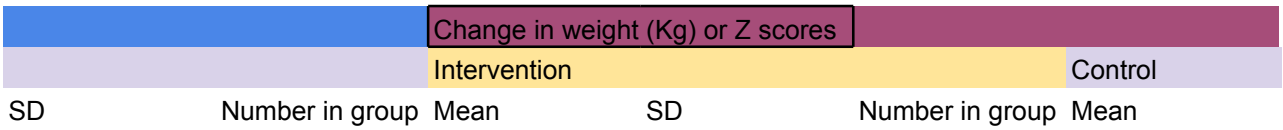
For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Kg				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Kg				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in weight (Kg) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

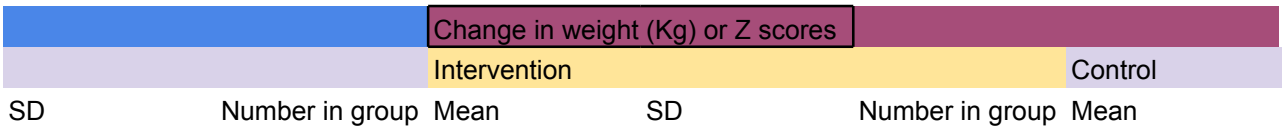
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Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Kg				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



For peer review only



SD Number in group

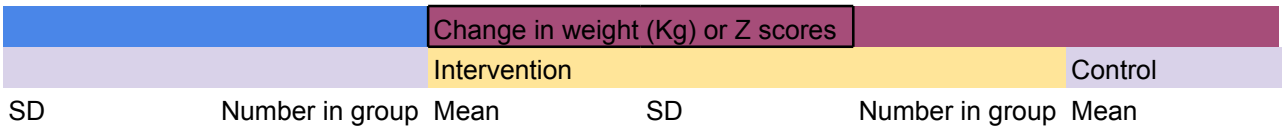
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Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Kg				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



For peer review only



SD Number in group

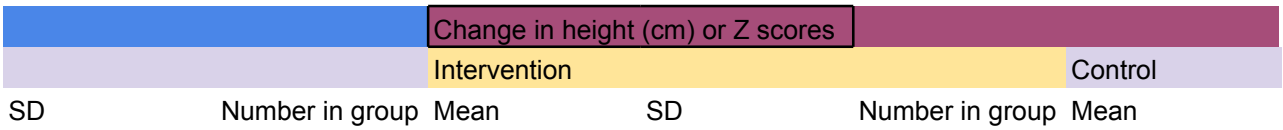
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Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



For peer review only



SD Number in group

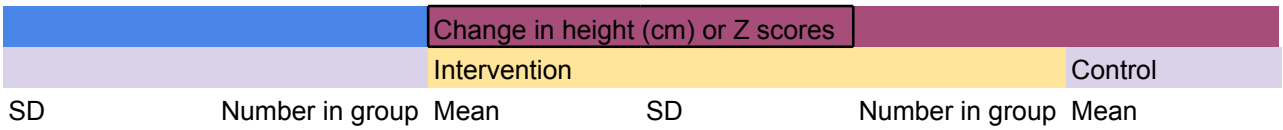
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Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in height (cm) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in height (cm) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in height (cm) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in height (cm) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in height (cm) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

For peer review only

Study ID (Last name and year)	Z scores			
	Intervention			Control
	Mean	SD	Number in group	Mean

For peer review only

	Centimeters				
	Intervention				Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only

		Change in height (cm) or Z scores			
		Intervention			Control
SD	Number in group	Mean	SD	Number in group	Mean

For peer review only



SD Number in group

For peer review only

year)	Iron deficiency anemia at 7 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Iron deficiency anemia at 9 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Iron deficiency anemia at 7 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Iron deficiency anemia at 7 months (yes/no)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Blood iron level at 7 months (continuous outcome)				
	in intervention	intervention	in control grouop	the control group	notes

For peer review only

year)	Blood iron level at 9 months (continuous outcome)				
	in intervention	intervention	in control grouop	the control group	notes

For peer review only

year)	Blood iron level at 12 months (continuous outcome)				
	in intervention	intervention	in control grouop	the control group	notes

For peer review only

year)	Blood iron level at longest follow up (continuous outcome)				
	in intervention	intervention	in control grouop	the control group	notes

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year)	Blood ferritin level at 7 months (continuous outcome)				
	level in	intervention	level in control	the control group	notes

For peer review only

year)	Blood ferritin level at 9 months (continuous outcome)				
	level in	intervention	level in control	the control group	notes

For peer review only

year)	Blood ferritin level at 12 months (continuous outcome)				
	level in	intervention	level in control	the control group	notes

For peer review only

year)	Blood ferritin level at longest follow up (continuous outcome)				
	level in	intervention	level in control	the control group	notes

For peer review only

year)	Hemoglobin concentration in the stool 7 months (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the stool 9 months (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the stool 12 months (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the stool longest follow up (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the serum 7 months (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the serum 9 months (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the serum 12 months (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Hemoglobin concentration in the serum longest follow up (continuous outcome)				
	concentration in	intervention	concentration	the control group	notes

For peer review only

year)	Fatty acid status at 7 months(continuous outcome)				
	intervention	intervention	group	the control group	notes

For peer review only

year)	Fatty acid status at 9 months (continuous outcome)				
	intervention	intervention	group	the control group	notes

For peer review only

year)	Fatty acid status at 12 months (continuous outcome)				
	intervention	intervention	group	the control group	notes

For peer review only

year)	Fatty acid status at longest follow up (continuous outcome)				
	intervention	intervention	group	the control group	notes

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year)	Diarrhea (> 3 loose stools per day) at 7 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

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year)	Diarrhea (> 3 loose stools per day) at 9 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Diarrhea (> 3 loose stools per day) at 12 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Diarrhea (> 3 loose stools per day) at longest follow up (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Constipation (<3 bowel movements per week) at 7 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Constipation (<3 bowel movements per week) at 9 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Pneumonia (as defined by authors) at 7 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Pneumonia (as defined by authors) at 9 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Pneumonia (as defined by authors) at 12 months (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Pneumonia (as defined by authors) at longest follow up (dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	outcome)				
	events in	intervention	events in control	the control group	notes

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year)	(dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	(dichotomous outcome)				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Obesity (dichotomous outcome) at 1 year				
	infants in	intervention	infants in control	the control group	notes

For peer review only

year)	Obesity (dichotomous outcome) at longest follow up				
	infants in	intervention	infants in control	the control group	notes

For peer review only

year)	Overweight (dichotomous outcome) at 1 year				
	infants in	intervention	infants in control	the control group	notes

For peer review only

year)	Overweight (dichotomous outcome) at longest follow up				
	infants in	intervention	infants in control	the control group	notes

For peer review only

year)	Neurodevelopmental outcomes (continuous outcome) at 1 year				
	events in	intervention	events in control	the control group	notes

For peer review only

year)	Neurodevelopmental outcomes (continuous outcome) at longest follow up				
	events in	intervention	events in control	the control group	notes

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with systematic review protocol submissions to BioMed Central journals from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

An Editorial from the Editors-in-Chief of *Systematic Reviews* details why this checklist was adapted - **Moher D, Stewart L & Shekelle P:** Implementing PRISMA-P: recommendations for prospective authors. *Systematic Reviews* 2016 **5**:15

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	48
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	86-87
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	9-42
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	401- 404
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☒	☐	377-380
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	410-411
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	410-411
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	405-408
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	102-121

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	123-126
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	129-140; 142-152
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	226-237
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	Appendix 1 (515-637)
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	254-264
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	240-252
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	282-309; 311-329
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	296- 309; 411
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	170-224
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	266-280
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	282-309
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	282-309; 331-342
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	348-356; 358-360
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	☒	n/a; quantitative

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
					analysis planned see 15a-c
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☒	☐	344-346
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	362-375

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BMJ Open

The effect of consumption of animal milk compared to infant formula for non-breastfed/mixed-fed infants 6-11 months of age. A systematic review (protocol)

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046370.R1
Article Type:	Protocol
Date Submitted by the Author:	22-Dec-2020
Complete List of Authors:	Imdad, Aamer; SUNY Upstate Medical University, Pediatric Gastroenterology Ehrlich, Julie; SUNY Upstate Medical University Catania, Joseph; SUNY Upstate Medical University Tanner-Smith, Emily; University of Oregon, Smith, Abigail; SUNY Upstate Medical University Tsistinas, Olivia; SUNY Upstate Medical University Bhutta, Zulfiqar; Aga Khan University, Division of Women and Child Health
Primary Subject Heading:	Paediatrics
Secondary Subject Heading:	Nutrition and metabolism, Paediatrics, Public health
Keywords:	NUTRITION & DIETETICS, Paediatric gastroenterology < PAEDIATRICS, PAEDIATRICS, Community child health < PAEDIATRICS, Anaemia < HAEMATOLOGY

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The effect of consumption of animal milk compared to infant formula for non-breastfed/mixed-fed infants 6-11 months of age. A systematic review (protocol)

Aamer Imdad, Julie Melissa Ehrlich, Joseph Catania, Emily Tanner Smith, Abigail Smith, Olivia Tsistinas, Zulfiqar Ahmed Bhutta

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Target Journal: "BMJ Open"

Type of study: Protocol

The effect of consumption of animal milk compared to infant formula for non-breastfed/mixed-fed infants 6-11 months of age. A systematic review (protocol)

Aamer Imdad, Julie Ehrlich, Joseph Catania, Emily Tanner Smith, Abigail Smith, Olivia Tsistinas, Zulfiqar Ahmed Bhutta

Abstract

Introduction

Prevalence rates of breastfeeding remain low even though the World Health Organization and American Academy of Pediatrics recommend exclusive breastfeeding for the first 6 months of life in combination with appropriate complementary feeding beyond six 6 months of age. There have been several studies that address the implication of drinking animal milk and/or infant formula on children's health and development when breastfeeding is not offered during the first year of life. Vast improvements have been made in infant formula design, which may increase its benefits compared to animal's milk. The objective of this review is therefore to synthesize the most recent evidence on the effects of the consumption of animal milk compared to infant formula in non-breastfed or mixed breastfed infants aged 6-11 months.

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Methods and Analysis

We will conduct a systematic review and meta-analysis of studies that assessed the effect of animal milk compared to formula or mixed-fed (breastmilk and formula) on infants aged 6-11 months. The primary outcomes of interest include anemia, gastrointestinal blood loss, weight for age, height for age, and weight for height. We will include randomized and non-randomized studies with a control group. We will use the Cochrane risk of bias tools to assess the risk of bias. We will use meta-analysis to pool findings if the identified studies are conceptually homogenous and data are available from more than one study. We will assess the overall quality of evidence using the GRADE approach.

Ethics and Dissemination

This is a systematic review, so no patients will be directly involved in the design or development of this study. The findings from this systematic review will be disseminated to relevant patient populations and caregivers and will guide the World Health Organization's recommendations on formula consumption versus animal milk in infants aged 6-11 months.

Registration

PROSPERO ID: CRD42020210925

Strengths and Limitations of the Study

- We will search several databases for relevant literature and will included randomized and non-randomized studies.
- We will conduct risk-of-bias for each outcome and use the GRADE to assess the overall quality of evidence.

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- We will conduct sub-group analysis for age group (7 months vs. 9 months), country, type of feeding (non-breastfed vs mixed fed) and type of animal milk.
- We will conduct meta-analyses if data are available from more than one study, and if there is clinical and methodological homogeneity in the included studies.
- All the included studies may not report the data for the infants 6-11 months of age which will be a limitation of the study.

Keywords: Animal's milk, infant formula, anemia

Introduction

The World Health Organization and American Academy of Pediatrics recommend exclusive breastfeeding for the first 6 months of life and continuation of breastfeeding beyond 6 months in combination with appropriate complementary feedings for up to 2 years or beyond[1,2]. However, prevalence rates of breastfeeding still remain low in many contexts, with many infants still being given formula milk or cow's milk before or after 6 months of age[3,4]. Use of cow's milk in infancy has been associated with gastrointestinal blood loss, iron deficiency anemia, and increased solute load for kidneys[5-8]. The composition of cow's milk is vastly different from that of breastmilk and infant formula, which is manufactured to mimic breastmilk[9]. Cow's milk has a high amount of protein (3.4 g/100 ml vs .7-1.4100 ml in breastmilk)[10], and the composition of the protein is different—50-80% of breastmilk is comprised of whey protein versus only 18% for cow's milk[9]. Ingesting a large amount of protein in the first year of life has been associated with childhood obesity[6,9]. Unmodified goat's milk also contains high amounts of protein and mineral content and low folate content. Goat's milk does not appear to have any clear nutritional

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advantage over cow's milk and is not less allergenic[11]. There have been several studies that have examined the implications of drinking animal milk and/or infant formula on children's health and development[5,7,12-17]. The past two decades have also yielded vast improvements in the design of infant formulas, which may increase the benefits of infant formula over animal milk[18]. The objective of the proposed review is, therefore, to synthesize the most recent research on the effects of the consumption of animal milk compared to infant formula in non-breastfed or mixed breastfed infants aged 6-11 months of age.

Objective

For non-breastfed or mixed-fed (breastmilk and formula) infants 6-11 months of age, is the consumption of animal milk compared to infant formula, associated with beneficial or adverse outcomes for health and development?

Methods and Analysis

Study type: We will include individual and cluster randomized trials. We will also include quasi-randomized experimental design studies, defined as those that attempted to randomly allocate participants to conditions but the method of randomization was not truly random[19]. For instance, an example of a quasi-randomized design is an experimental study where the randomization was attempted but the methods of sequence generation were inadequate such as randomization based on date of birth of participants, date of enrollment in the study. We will also include cohort studies, defined as a study on a group of participants based on certain criteria (e.g., geography, age, etc.) but none of the participants have the outcome at the time of recruitment and develop outcomes over time in relation to exposure vs. no exposure to a risk factor. A cohort study could

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be prospective or retrospective[19]. We will exclude case control studies, case series, case reports and cross-sectional studies. All analyses will be conducted and reported separately for randomized and non-randomized studies.

Population: We will include studies that included apparently healthy infants 6-11 months of age who were non-breastfed or mixed fed (breastmilk and formula) irrespective of gestational age and birthweight. We will include studies that started animal milk feeding between 4 months and 11 months of age. If a study includes participants who began intervention before 4 months of age but includes participants older than 4 months of age as well, we will consider that study eligible if more than half (50%) of the study participants meet the inclusion criteria (i.e. 4-11 months of age). We will include studies where participants are diagnosed with HIV or have had HIV exposure. We will exclude studies if participants are diagnosed with AIDS. We will exclude studies that focused on breastfed infants only without other types of milk provided. We will exclude studies with participants who have chronic diseases such as bronchopulmonary dysplasia, genetic disorders, aerodigestive problems, or congenital anomalies.

***Intervention*:** We will include studies that assessed the effect of animal milk in infants 6-11 months of age. The animal milk could be cow milk, goat milk, or camel milk. We will include studies in which animal milk was the main milk drink as defined by study authors or more than 50% of the infant's milk intake was animal milk. We will include studies irrespective if the animal milk was boiled or not boiled, pasteurized or unpasteurized, or if the animal milk was full-fat, reduced fat, or skim milk. We will exclude studies that used plant-based milk such as rice milk, pea milk, coconut milk, or almond milk.

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***Comparison*:** The comparison group in the included studies will be formula feeding or mixed feeding (i.e., breastfeeding and formula feeding). We will include studies irrespective of the type of formula used; namely, this could include cow's milk-based formula, partially or extensively hydrolyzed formula, or plant-based formulas such as soy formula. The Federal Food, Drug, and Cosmetic Act (FFDCA) defines infant formula as "a food which purports to be or is represented for special dietary use solely as a food for infants by reason of its simulation of human milk or its suitability as a complete or partial substitute for human milk"[20].

Outcomes:

Primary outcomes

-Anemia (dichotomous outcome, as defined by authors) at 7 months, 9 months, 12 months, and the longest follow up

-Gastrointestinal blood loss (dichotomous outcome based on stool occult testing) at 7 months, 9 months, 12 months, and the longest follow up

-Weight for age (kg or Z scores) at 7 months, 9 months, 12 months, and the longest follow up

-Height for age (cm or Z scores) at 7 months, 9 months, 12 months, and the longest follow up

-Weight for height Z score at 7 months, 9 months, 12 months, and the longest follow up

Secondary outcomes

-Iron deficiency anemia (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

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-Blood iron level (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Blood ferritin level (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Hemoglobin concentration in the stool (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Hemoglobin concentration in the serum (continuous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Fatty acid status (continuous outcome) at 7 months, 9 months, 12 months and longest follow up

-Gut health: Diarrhea (> 3 loose stools per day) (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Gut health: Constipation (<3 bowel movements per week) (dichotomous outcome) at 7 months, 9 months, 12 months and longest follow up

-Pneumonia (as defined by authors) (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

-Allergy (IgE-Mediated and non-IgE Mediated and mixed) (dichotomous outcome) at 7 months, 9 months, 12 months, and the longest follow up

We will include both immunoglobulin E (IgE) mediated, non-IgE mediated and mixed milk allergies. IgE mediated can be identified with a skin prick test or blood test for specific IgE and symptoms typically arise immediately after ingesting the milk product. Non-IgE mediated allergies will not be identified with a skin prick test or blood test and symptoms typically arise 48 hours after ingestion of milk product [21]. Non-IgE mediated allergies can be identified using

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allergen elimination and standard challenge procedure in which the allergen is removed from the diet for up to 2 weeks and then reintroduced to determine if symptoms are elicited[22].

- Obesity (dichotomous outcome) at 1 year and longest follow up

Obesity is defined as a BMI at or above the 95th percentile for children and teens of the same age and sex[23]. It can also be defined as BMIZ>3 for children under 5 years of age.

-Overweight (dichotomous outcome) at 1 year and the longest follow up

Overweight is defined as a BMI at or above the 85th percentile and below the 95th percentile for children and teens of the same age and sex[23]. It can also be defined as BMIZ>2 for children under 5 years of age.

-Neurodevelopmental outcomes (continuous outcome) at 1 year and the longest follow up

The term neurodevelopment is a composite term that refers to cognitive, neurologic, and/or sensory outcomes. This may include intellectual disability as measured on the Mental Developmental Index of the Bayley Scales of Infant Development, gross motor delay measured on Gross Motor Function Classification System, etc.

If a study does not report outcomes at 7 months but instead reports outcomes at 6 months or 8 months, it will be included with outcomes at 7 months. Similarly, if a study does not report outcomes at 9 months but instead reports outcomes at 10 months, it will be included with outcomes at 9 months. If a study does not report outcomes at 12 months, but instead reports those outcomes at 11, 13, or 14 months, it will be included with outcomes at 12 months.

Literature Search:

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We will conduct systematic electronic searches on multiple databases, including PubMed, EMBASE, the Cochrane Central Register for Controlled Trials, Web of Science, CINHAL, Scopus, WHO Global Index Medicus. There will be no restrictions applied to the searches based on outcomes, study design, publication status, publication date, or language. A proposed search strategy for PubMed is shown in Appendix 1. We will search ClinicalTrials.gov to identify ongoing studies. We will also search the websites of relevant international agencies, such as the WHO (including WHO's Reproductive Health Library), UNICEF, Global Alliance for Improved Nutrition, International Food Policy Research Institute, International Initiative for Impact Evaluation (3ie), Nutrition International, World Bank, USAID and affiliates (e.g., FANTA, SPRING), and the World Food Program. The reference sections of previously published reviews and the latest published studies will also be searched for potentially eligible studies.

Data Extraction and Synthesis:

Selection of studies

Searches from all the databases will be combined in bibliographic software (EndNote) and duplicates will be removed. Two authors (AI and JC or AI and JE) will first screen the titles to identify potentially eligible studies; any studies retained at this step will proceed to the second stage of screening for a full-text review; finally, any studies deemed eligible during the full-text review will proceed to the third step of full data extraction. Any conflict will be resolve by discussion and with the help of the senior author on the team if needed. We will use coding software (Covidence) to conduct the screening[24]. If a study is only available in abstract form, we will write to authors to obtain details on methods and results. If a study is available in a language other than English, we will attempt to have the translation completed using the local resources. If

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a study was published in more than one report (multiple publications), we will count those multiple reports as a single study and extract information from all the available reports as needed.

Data Extraction

A data extraction sheet will be designed and piloted for the extraction of information from the selected studies (supplementary file). Two authors (JE and JC) will independently extract the data and compare their findings. Any conflict will be resolved by discussion and with the help of the senior author on the team if needed. We will extract the information on type of study (randomized control trial, quasi-randomized experimental design, or cohort study) study site, study year, study population (age, nutritional status, percent female), methods for data collection and analysis, intervention (dose, duration, frequency), age of outcome assessment, comparison, outcomes, whether the results were adjusted for confounders, what confounders were considered (socioeconomic status, maternal employment, maternal age, maternal level of education, age-appropriate complementary feeding) and risk of bias (supplementary document).

Assessment of risk of bias in included studies

Study risk of bias will be assessed using the Cochrane risk of bias tool (ROB 2.0)[25] for randomized controlled trials and using the Cochrane risk of bias in non-randomized studies (ROBINS-I) tool for non-randomized studies of interventions[26]. Two review authors will independently evaluate and agree on the risk of bias for the individual studies for an outcome. Any disagreements will be resolved by discussion and if no agreement can be made, the senior review author will be consulted. We will assess the effect of assignment to intervention (the

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intention-to-treat effect) by addressing five domains of signaling questions including: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result. Each domain will get a ranking of low risk of bias, some concerns of bias, or high risk of bias. We will include quotes from the study for each signaling question as evidence for our ranking decision. The overall risk of bias will be determined based on the worst ranking for individual domains. For example, if only one domain is ranked "some concerns" then the overall risk of bias will be "some concerns".

Data synthesis

We will develop a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram based on the search strategy and eligibility assessment to show the flow of included and excluded studies[27]. We will analyze the randomized and non-randomized studies separately. We will report findings from all included studies in a narrative synthesis and will also conduct meta-analyses to synthesize evidence across studies quantitatively. Meta-analyses will be conducted when data are available from more than one study, and when there is clinical and methodological homogeneity in the included studies. Dichotomous outcomes will be measured using relative risk effect sizes and will be reported with their corresponding 95 % confidence intervals. Continuous outcomes will be measured using mean difference effect sizes and will also be reported along with their 95 % confidence intervals. We will use the generic inverse variance weighting method for meta-analysis. We will use the random effects model for meta-analysis given that there might be heterogeneity in effects due to variability in the study populations and interventions used. We will use RevMan[28] and Stata[29] software for the statistical analysis.

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In the case of morbidity outcomes, we will combine all available data whenever possible if outcomes are measured in different ways. For example, we will include all types of diarrhea (mild, moderate and severe) as a dichotomous value (yes/no) if participants had greater than 3 instances of loose stools per day. We will include occurrence of anemia, gastrointestinal blood loss based on stool occult testing, iron deficiency anemia, pneumonia, and allergies (non-IgE, IgE mediated) throughout the study as dichotomous values (yes/no). For dichotomous outcomes, we will extract the total number of participants in each group and the number of participants experiencing an event. We will pool the dichotomous outcomes in a meta-analysis to obtain a summary estimate in the form of relative risk and report it with its corresponding 95 % confidence interval (CI). We will include hemoglobin concentration in stool, weight for age, height for age, and weight for height as continuous values. For continuous outcomes, we will pool the data to obtain a pool mean difference and report it with its 95 % CI. In the event that data are reported in different units (e.g. few studies report weight in kg and the others report in Z scores), we will use a standardized mean difference effect size and report it with its 95 % CI.

Unit of analysis issues

Multiple-arm trial

Studies with multiple treatment arms will be included if eligible. For multiple-arm trials we will include data in a way that the only difference between the groups is use of animal milk.

Cluster trials

Cluster assignment trials will be analyzed together with individual randomized trials. We will use the cluster adjusted values; If the trial results are not adjusted for cluster design, we will adjust the result by methods given in the Cochrane handbook for systematic reviews[19].

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Studies with missing data

We will document attrition during data extraction. If the authors report the missing data and report the imputations for the missing data, we will use the latter. If data are missing for some cases, or if reasons for dropout are not reported, we will contact the trial authors to request the full data. In the event that a study does not report the SD for the continuous outcome, we will write to authors to request the data. If the SD data are not available from authors, we will use SD from a similar study that has similar study population. We will prefer to use the final values of a continuous outcome for a given follow up. If the final values are not available but the difference between the end and the start of the study, we will write to authors to request the final values. If the final values are not available, we will use the difference or rate of change.

Assessment of heterogeneity

Assessment of heterogeneity in meta-analysis determines how different the included studies are from each other. Clinical heterogeneity refers to differences in patient population or treatment whereas methodological heterogeneity refers to differences in how the study was designed. Statistical heterogeneity refers to large differences in the outcome (outside of chance) that may be due to clinical or methodological heterogeneity[30]. Statistical heterogeneity will be assessed using Tau^2, χ^2 and I^2 statistics. We will assess statistical heterogeneity by visual inspection of forest plots, by performing the Chi^2 test (assessing the P value) and by calculating the Tau^2 and I^2 statistic. Statistical heterogeneity will be considered substantial if the P value is less than 0.10, I^2 value exceeds 50%, and inspection of forest plots show substantial variability in the effect of

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the intervention. We will perform subgroup analysis to determine the reasons for any identified statistical heterogeneity.

Assessment of reporting bias

Small study and publication bias will be assessed using funnel plots and regression tests for funnel plot asymmetry when the meta-analysis includes at least ten studies.

Subgroup analysis and investigation of heterogeneity

-Age group: 7 months vs. 9 months

If a study does not begin intervention at 7 months but instead at 6 months or 8 months, it will be included in the 7 months age sub-group. Similarly, if a study does not begin intervention at 9 months but instead at 10 months, it will be included with outcomes at 9 months.

-Country: Low and middle-income country vs. high-income country

-Type of Feeding: Non-breastfeeding vs. mixed feeding

- Type of Animal Milk: Cow, goat, buffalo, camel, sheep

We will test the subgroups difference by using chi-square test.

Sensitivity analysis

-Studies with a high overall risk of bias excluded

-Random vs. fixed effect meta-analysis model

Rating of overall quality of evidence

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We will assess the overall quality of evidence for the effect of the intervention on each primary outcome using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) using the software GradePro[31]. The GRADE approach entails identifying and assessing many different features that affect the certainty of evidence in the review, including type of study design, statistical heterogeneity, directness of evidence, within-study risk of bias, risk of publication bias, and precision of effect estimates[32]. We will consider evidence from randomized trials as high quality and downgrade the evidence one level for serious limitations (table 1).

Table 1: ***GRADE method for rating the quality of evidence***

Study Design	**Quality of Evidence**	**Lower if**	**Higher if**
Randomized trial	High	Risk of Bias -1 Serious -2 Very serious Inconsistency -1 Serious -2 Very serious Indirectness -1 Serious -2 Very serious Imprecision -1 Serious -2 Very serious Publication bias -1 Serious -2 Very serious	Large effect + 1 Serious + 2 Very serious Dose response + 1 Evidence of a gradient All plausible confounding + 1 Would reduce a demonstrated effect or + 1 Would suggest a spurious effect when results show no effect
Quasi-Randomized trial	Moderate		
	Low		
	Very low		

We will rate the overall quality of the body of evidence as very low (we have very little confidence in the effect estimate), low (we have limited confidence in the effect estimate),

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moderate (we have moderate confidence in the effect estimate; the true effect is likely close to the estimate of the effect), or high (we have high confidence that the true effect lies close to that of the estimate of the effect). We will present the results of the GRADE assessment in the form of a summary of findings tables for the primary outcomes.

Amendments

We will do the literature searches, screening of titles, selection of studies, data extraction, and analysis according to a priori plan described in this protocol. If we do any additional analysis or change any of the a priori strategies, we will clearly describe that in the methods section.

Patient and Public Involvement

No patient or public involvement was considered in preparation of this protocol.

List of abbreviations

World Health Organization (WHO)

The Federal Food, Drug, and Cosmetic Act (FFDCA)

Confidence Interval (CI)

Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

Grading of Recommendations Assessment, Development, and Evaluation (GRADE)

Acquired immunodeficiency syndrome (AIDS)

Consent for publication

Authors give consent for publication of this review.

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Availability of data and materials

We will keep all the data available for review by the editors and peer reviewers and will provide raw data for the general public on request.

Ethics and Dissemination

This is a systematic review and there will be no direct contact with the patients. The findings from this systematic review will be disseminated to relevant patient populations and caregivers as they will be used to guide recommendations through the World Health Organization on whether the use of infant formula is better than animal's milk in infants aged 6-11 months.

Authors' contributions

AI and JE and JC wrote the first draft of the manuscript. AS and OT developed the search strategy. ETS provided the support for methodology. ZAB provided the support for the content being evaluated in the review. All authors edited and reviewed the manuscript.

Acknowledgements

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Funding Statement:

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Competing Interests Statement

Authors declare they do not have any conflicts of interest to disclose.

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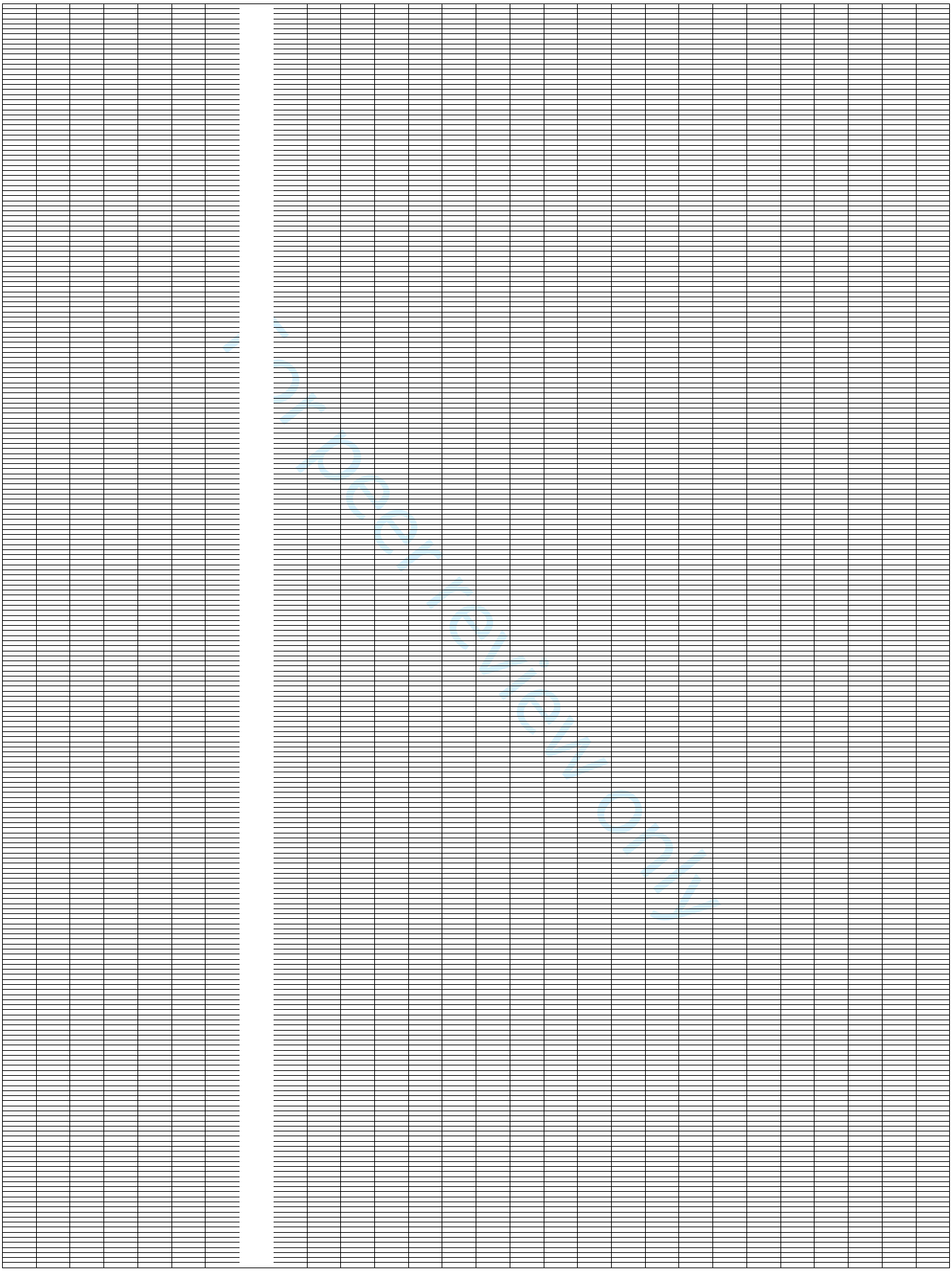
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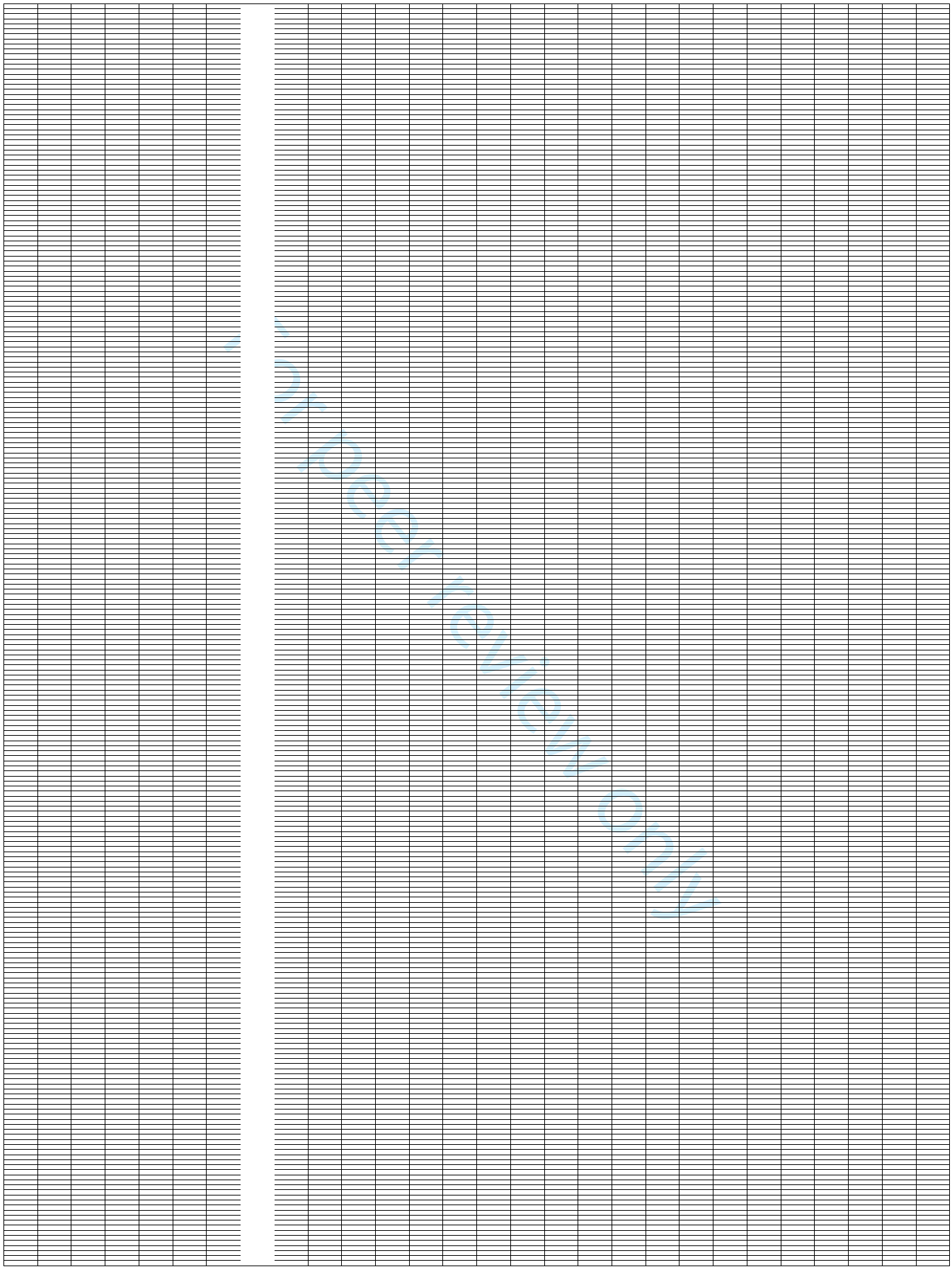
Author	Publication year	Name of data extractor	Form: conference abstract only (1), journal publication (2),	Study Design	Cluster Design Trial: no (0), yes (1)	Study Setting			Population								Characteristic of Intervention			Characteristic of comparison			Study confounders		Notes	Minimal loss to follow up? (more than 20% is serious limitation)
						Country	LMIC (1) vs. HIC (2) --> check	Year(s) of Data Collection	Description of Participants		Age initiation of animal milk	Nutritional Status: healthy (1),	% Female (N) (Give for total	Were there more than one eligible	Total number of eligible	Total number of eligible	Number of Participants	Type of Animal milk (eg: cow milk,	Frequency of animal milk	Volume of animal milk feedings	Type: breastmilk and formula	Volume	Frequency	Were results adjusted for	What confounders		
									Inclusion Criteria	Exclusion Criteria																

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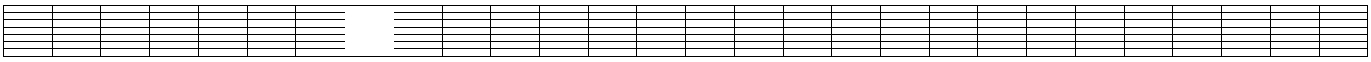
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Appendix 1:

PubMed

(((("cow milk*"[tiab] OR "cow's milk*"[tiab] OR "cows milk*"[tiab] OR "bovine milk*"[tiab] OR "bovine's milk*"[tiab] OR "goat milk*"[tiab] OR "goat's milk*"[tiab] OR "goats milk*"[tiab] OR "caprine milk*"[tiab] OR "buffalo milk*"[tiab] OR "buffalo's milk*"[tiab] OR "camel milk*"[tiab] OR "camel's milk*"[tiab]OR "camels milk*"[tiab] OR "sheep milk*"[tiab] OR "sheep's milk*"[tiab] OR "ewe milk*"[tiab] OR "ewe's milk*"[tiab] OR "ewes milk*"[tiab] OR "ovine milk*"[tiab]) AND ("Infant Formula"[Mesh] OR formula*[tiab])) AND ("Infant"[Mesh] OR infant*[tiab] OR infancy[tiab] OR baby[tiab] OR babies[tiab] OR neonat*[tiab] OR "neo nat*"[tiab] OR newborn*[tiab] OR "new born*"[tiab] OR "newly born*"[tiab] OR "Child"[Mesh] OR child*[tiab] OR youth[tiab] OR juvenile*[tiab])) NOT ("Animals"[Mesh] NOT ("Animals"[Mesh] AND "Humans"[Mesh]))

CINAHL

(TI ("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*") OR AB ("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel's milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*"OR "ewe milk*" OR "ewe's milk*" OR "ewes milk*" OR "ovine milk*"))
AND
((MH "Infant Formula") OR (TI formula* OR AB formula*))
AND
(((MH "Child") OR (MH "Infant+")) OR (TI (infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*) OR AB (infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)))
NOT
(((MH "Animals+") OR (MH "Animal Studies") OR (TI "animal model*")) NOT (MH "human"))
Exclude Medline Records

SCOPUS

(TITLE-ABS ("cow milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camels milk*" OR "sheep milk*" OR "sheep's milk*" OR "ewe milk*" OR OR "ewes milk*" OR "ovine milk*")) AND (TITLE-ABS (formula*)) AND (TITLE-ABS (infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)) AND NOT INDEX (medline)

Embase

1 'cow milk'/exp OR 'goat milk'/exp OR 'buffalo milk'/exp OR 'camel milk'/exp OR 'sheep milk'/exp
2 'cow milk*':ti,ab OR 'cows milk*':ti,ab OR 'cow s milk*':ti,ab OR 'bovine milk*':ti,ab OR 'bovine s milk*':ti,ab OR 'goat milk*':ti,ab OR 'goat s milk*':ti,ab OR 'goats milk*':ti,ab OR 'caprine milk*':ti,ab OR 'buffalo milk*':ti,ab OR 'buffalo s milk*':ti,ab OR 'camel milk*':ti,ab OR 'camel s milk*':ti,ab OR 'camels milk*':ti,ab OR 'sheep milk*':ti,ab OR 'sheep s milk*':ti,ab OR 'ewe milk*':ti,ab OR 'ewes milk*':ti,ab OR 'ewe s milk*':ti,ab OR 'ovine milk*':ti,ab
3 #1 OR #2
4 'artificial milk'/exp
5 formula*:ti,ab
6 #4 OR #5
7 'infant'/exp OR 'infancy'/exp OR 'baby'/exp OR 'newborn'/exp OR 'child'/exp OR 'juvenile'/exp
8 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR neonat*:ti,ab OR 'neo nat*':ti,ab OR newborn*:ti,ab OR 'new born*':ti,ab OR 'newly born*':ti,ab OR child*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
9 #7 OR #8
10 #3 AND #6 AND #9
11 #10 NOT ([animals]/lim NOT [humans]/lim)

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12 #11 NOT [medline]/lim

Cochrane CENTRAL

1 “cow milk*”:ti,ab OR “cows milk*”:ti,ab OR “cow’s milk*”:ti,ab OR "bovine milk*":ti,ab OR "bovine's milk*":ti,ab OR “goat milk*”:ti,ab OR “goat’s milk*”:ti,ab OR “goats milk*”:ti,ab OR "caprine milk*":ti,ab OR “buffalo milk*”:ti,ab OR “buffalo’s milk*”:ti,ab OR “camel milk*”:ti,ab OR “camel’s milk*”:ti,ab OR “camels milk*”:ti,ab OR “sheep milk*”:ti,ab OR “sheep’s milk*”:ti,ab OR "ewe milk*":ti,ab OR "ewes milk*":ti,ab OR "ewe's milk*":ti,ab OR "ovine milk*":ti,ab
2 MeSH descriptor: [Infant Formula] explode all trees
3 formula*:ti,ab
4 #2 OR #3
5 MeSH descriptor: [Child] explode all trees
6 MeSH descriptor: [Infant] explode all trees
7 infant*:ti,ab OR infancy:ti,ab OR baby:ti,ab OR babies:ti,ab OR neonat*:ti,ab OR “neo nat*”:ti,ab OR newborn*:ti,ab OR “new born*”:ti,ab OR “newly born*”:ti,ab OR child*:ti,ab OR youth:ti,ab OR juvenile*:ti,ab
8 #5 OR #6 OR #7
9 MeSH descriptor: [Animals] explode all trees
10 MeSH descriptor: [Humans] explode all trees
11 (#9 NOT(#9 AND #10))
12 #1 AND #4 AND #8 NOT #11
13 "accession number" near pubmed
14 #12 NOT #13

LILACS

ti:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel’s milk*" OR "camels milk*" OR "sheep milk*" OR "sheep’s milk*" OR "ewe milk*" OR "ewe’s milk*" OR "ewes milk*" OR "ovine milk*") OR (ab:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel’s milk*" OR "camels milk*" OR "sheep milk*" OR "sheep’s milk*" OR "ewe milk*" OR "ewe’s milk*" OR "ewes milk*" OR "ovine milk*")) AND (((ti:(formula*)) OR (ab:(formula*))) OR ((mh:("infant formula")))) AND (((ti:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)) OR (ab:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*))) OR ((mh:("infant")) OR ((mh:("child"))))) AND NOT ((mh:("animals")) AND NOT ((mh:("animals")) AND ((mh:("humans")))))

Global Index Medicus

ti:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel’s milk*" OR "camels milk*" OR "sheep milk*" OR "sheep’s milk*" OR "ewe milk*" OR "ewe’s milk*" OR "ewes milk*" OR "ovine milk*") OR (ab:("cow milk*" OR "cow's milk*" OR "cows milk*" OR "bovine milk*" OR "bovine's milk*" OR "goat milk*" OR "goat's milk*" OR "goats milk*" OR "caprine milk*" OR "buffalo milk*" OR "buffalo's milk*" OR "camel milk*" OR "camel’s milk*" OR "camels milk*" OR "sheep milk*" OR "sheep’s milk*" OR "ewe milk*" OR "ewe’s milk*" OR "ewes milk*" OR "ovine milk*")) AND (((ti:(formula*)) OR (ab:(formula*))) OR ((mh:("infant formula")))) AND (((ti:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*)) OR (ab:(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new born*" OR "newly born*" OR child* OR youth OR juvenile*))) OR ((mh:("infant")) OR ((mh:("child"))))) AND NOT ((mh:("animals")) AND NOT ((mh:("animals")) AND ((mh:("humans")))))AND (db:("WPRIM" OR "IMEMR" OR "IMSEAR" OR "AIM"))

Web of Science

1 TI=(“cow* milk*" OR "bovine* milk*" OR "goat* milk*" OR "caprine milk*" OR "buffalo* milk*" OR “camel* milk*" OR "sheep* milk*" OR “ewe* milk*” OR "ovine milk*")

2 AB=(“cow* milk*" OR "bovine* milk*" OR "goat* milk*" OR "caprine milk*" OR "buffalo* milk*" OR "camel* milk*" OR "sheep* milk*" OR “ewe* milk*” OR "ovine milk*")
3 #2 OR #1
4 TI=(formula*)
5 AB=(formula*)
6 #5 OR #4
7 TI=(infant* OR infancy OR baby OR babies OR neonat* OR "neo nat*" OR newborn* OR "new* born*" OR child* OR youth OR juvenile*)
8
9 #8 OR #7

10 #9 AND #6 AND #3

11 #9 AND #6 AND #3 Exclude Medline

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with systematic review protocol submissions to BioMed Central journals from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

An Editorial from the Editors-in-Chief of *Systematic Reviews* details why this checklist was adapted - Moher D, Stewart L & Shekelle P: Implementing PRISMA-P: recommendations for prospective authors. *Systematic Reviews* 2016 **5**:15

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	48
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	86-87
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	9-42
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	401- 404
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☒	☐	377-380
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	410-411
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	410-411
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	405-408
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	102-121

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	123-126
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	129-140; 142-152
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	226-237
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	Appendix 1 (515-637)
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	254-264
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	240-252
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	282-309; 311-329
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	296- 309; 411
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	170-224
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	266-280
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	282-309
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	282-309; 331-342

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	348-356; 358-360
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☐	☒	n/a; quantitative analysis planned see 15a-c
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☒	☐	344-346
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	362-375

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